Online Supplementary Material

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Appendix C to "On iterated Lorenz curves with applications: the multivariate case"

Abstract: This file contains Appendix C, which is provided as supplementary material.

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Appendix C
Appendix C.1

This appendix is dedicated to the analysis of the iterative equation (74). We investigate properties of this equation that enable us to establish its convergence and to approximate its limit. While the technical analysis herein is self-contained, several of the ideas and techniques will also be of service in the subsequent *Appendix D*.

Appendix C.1.1
L_n dynamics and fixed points

We start with some remarks on the notation. For convenience we denote below any of the marginal distributions $L^n_{1-}(x)$ or $L^n_{2-}(x)$ by $L_n(x)$ since the properties proved hold for both of them. Thus, our setting becomes

$$L_{n+1}(x) = \frac{\int_0^x L^{n,-1}(u)\left(1 - L^{n,-1}(u)\right)du}{\int_0^1 L^{n,-1}(u)\left(1 - L^{n,-1}(u)\right)du}, \textit{with } L_0(x) = \frac{\int_0^x F^{-1}(u)\left(1 - F^{-1}(u)\right)du}{\int_0^1 F^{-1}(u)\left(1 - F^{-1}(u)\right)du}. \tag{222}$$

For brevity we will sometimes write

$$w(x) = x(1-x), \qquad x \in [0,1], \tag{223}$$

and

$$T_n(x) = L_n^{-1}(x), \qquad x \in [0,1]. \tag{224}$$

We now turn to structural properties of the iterates L_n themselves. The first key step is a single-crossing property.

Lemma 60. *For each $n \geq 0$, the function L_n has two fixed points at 0 and 1, and a unique fixed point in $(0,1)$. If we denote the latter by c_n, then*

$$L_n(x) < x \quad \textit{for } x \in (0, c_n), \qquad L_n(x) > x \quad \textit{for } x \in (c_n, 1). \tag{225}$$

Moreover, there exists a closed interval $O_n \subset (0,1)$ such that $c_n \in O_n$ and

$$L'_n(x) \geq 1 \quad \textit{for all } x \in O_n. \tag{226}$$

Proof. Introducing shorthand notation, we may write (222) as

$$L_{n+1}(x) = \frac{\int_0^x w[T_n(u)]du}{\int_0^1 w[T_n(u)]du}, \tag{227}$$

where $T_n(x) = L_n^{-1}(x)$ and $w(x) = x(1-x)$. Clearly, $T_n(x) : [0,1] \to [0,1]$ and it is strictly increasing in $[0,1]$. Also, $w(x) : [0,1] \to [0, \frac{1}{4}]$ and it is strictly increasing in $[0, \frac{1}{2}]$ and strictly decreasing in $[\frac{1}{2}, 1]$, with a unique global maximum at $x = \frac{1}{2}$.

Let $h_n(x) = x - L_n(x)$. Differentiating h_{n+1} and applying the *Lagrange's Mean Value Theorem* in its integral form, yields

$$h'_{n+1}(x) = 1 - \frac{w[T_n(x)]}{\int_0^1 w[T_n(u)]du} = \frac{w[T_n(\xi_n)] - w[T_n(x)]}{\int_0^1 w[T_n(u)]du}, \tag{228}$$

where $\xi_n \in (0,1)$ is chosen so that

$$I_n = \int_0^1 w[T_n(u)]du = w[T_n(\xi_n)]. \tag{229}$$

The sign of $h'_{n+1}(x)$ is the sign of

$$w(T_n(\xi_n)) - w(T_n(x)) = [T_n(x) - T_n(\xi_n)]\,[T_n(x) + T_n(\xi_n) - 1]\,. \tag{230}$$

Because T_n is strictly increasing and w is unimodal with peak at $\frac{1}{2}$, there exist two points $c^{(1)}_{n+1}$ and $c^{(2)}_{n+1}$ in $(0,1)$ such that h_{n+1} is

$$\text{increasing on } [0, c^{(1)}_{n+1}], \quad \text{decreasing on } [c^{(1)}_{n+1}, c^{(2)}_{n+1}], \quad \text{increasing on } [c^{(2)}_{n+1}, 1], \tag{231}$$

with

$$c^{(1)}_{n+1} = \xi_n < L_n[1 - T_n(\xi_n)] = c^{(2)}_{n+1} \quad \text{when } \xi_n < L_n(\tfrac{1}{2}), \tag{232}$$

and

$$c^{(1)}_{n+1} = L_n[1 - T_n(\xi_n)] < \xi_n = c^{(2)}_{n+1} \quad \text{when } \xi_n > L_n(\tfrac{1}{2}). \tag{233}$$

We cannot have $c^{(1)}_{n+1} = c^{(2)}_{n+1}$ since by the *Mean Value Theorem* and the fact that $w(x)$ attains a maximum of $\frac{1}{4}$ that would imply a degenerate distribution for $L_n(x)$ (forced to be a constant), which cannot happen even if F is such due to the integration in (227). Since $h_{n+1}(0) = 0$ and $h_{n+1}(1) = 0$, the just derived monotonicity pattern of h_{n+1} by the *Bolzano's Intermediate Value Theorem* leads to the existence of a unique zero of $h_{n+1}(x)$ in $(c^{(1)}_{n+1}, c^{(2)}_{n+1})$; we denote this unique interior zero by c_{n+1}. This yields the claimed single-crossing property.

Differentiating once more we find

$$h''_{n+1}(x) = \frac{2L^{n,-1}(x) - 1}{(L^n)'(L^{n,-1}(x)) \displaystyle\int_0^1 w(T_n(u))du}. \tag{234}$$

Thus h'_{n+1} starts at 1 at $x = 0$, decreases on $[0, L_n(\frac{1}{2})]$, then increases on $[L_n(\frac{1}{2}), 1]$, with

$$h'_{n+1}(L_n(\tfrac{1}{2})) = 1 - \frac{1}{4\int_0^1 w(T_n(u))\,du} < 0, \tag{235}$$

and returns to 1 at $x = 1$. On each of the intervals

$$(0, L_n(\tfrac{1}{2})), \qquad (L_n(\tfrac{1}{2}), 1), \tag{236}$$

it attains the zeros $c^{(1)}_{n+1}$ and $c^{(2)}_{n+1}$ found above. Hence

$$h'_{n+1}(x) \le 0 \quad \text{for } x \in [c^{(1)}_{n+1}, c^{(2)}_{n+1}], \tag{237}$$

which is equivalent to

$$L'_{n+1}(x) \ge 1 \quad \text{for } x \in [c^{(1)}_{n+1}, c^{(2)}_{n+1}]. \tag{238}$$

Hence the interval O_n is $[c^{(1)}_n, c^{(2)}_n]$. □

The proof of the preceding lemma also yields the following immediate corollary, which will be essential for our subsequent analysis.

Corollary 61. *The inequality $L'_n(x) > 1$ holds for all x in the interval $(c^{(1)}_n, c^{(2)}_n)$.*

The next step is to show that the family $(L_n)_{n\ge 0}$ is equicontinuous and uniformly bounded on $[0,1]$, which follows from an explicit derivative bound. We first prove this claim under the assumption that the crossing points are uniformly bounded, and subsequently relax this restriction.

Claim 62. *Assume, in addition to the crossing pattern in Lemma 60, that the interior crossing points stay away from the endpoints, i.e. that there exists $\varepsilon > 0$ such that*

$$c_n \in [\varepsilon, 1-\varepsilon] \qquad \textit{for all } n \ge 0. \tag{239}$$

Then there exists a constant $M < \infty$, independent of n, such that

$$0 \le L_n'(x) \le M \qquad \text{for all } x \in [0,1],\ n \ge 0. \tag{240}$$

Consequently, each L_n is M-Lipschitz:

$$|L_n(x) - L_n(y)| \le M|x-y| \qquad \text{for all } x, y \in [0,1],\ n \ge 0. \tag{241}$$

Proof. Let

$$w(y) = y(1-y). \tag{242}$$

Since $T_n(x) \in [0,1]$ and w attains its maximum $1/4$ on $[0,1]$, we have

$$0 \le w(T_n(x)) \le \frac{1}{4} \qquad \text{for all } x \in [0,1],\ n \ge 0. \tag{243}$$

We first prove that the normalizing constants

$$I_n = \int_0^1 w(T_n(u))\, du \tag{244}$$

are uniformly bounded away from zero. Suppose, to the contrary, that this is not the case. Then there exists a subsequence, still denoted by n_k, such that

$$I_{n_k} \to 0. \tag{245}$$

Since each L_n is a distribution function on $[0,1]$, *Helly's selection principle* (see [5, Chapter 5] and [41, Chapter 36.5]) gives a further subsequence, again denoted by L_{n_k}, and a distribution function L^* such that

$$L_{n_k}(x) \to L^*(x) \tag{246}$$

at every continuity point x of L^*.

We now identify the possible form of L^*. Since

$$I_{n_k} = \int_0^1 T_{n_k}(u)(1 - T_{n_k}(u))\, du \to 0 \tag{247}$$

and

$$0 \le T_{n_k}(u)(1 - T_{n_k}(u)) \le \frac{1}{4}, \tag{248}$$

we may pass to a further subsequence such that

$$T_{n_k}(u)(1 - T_{n_k}(u)) \to 0 \qquad \text{for a.e. } u \in (0,1). \tag{249}$$

Thus every a.e. limit of the increasing functions T_{n_k} takes only the values 0 and 1. Since an increasing $\{0,1\}$-valued function is a threshold function, there exists $p \in [0,1]$ such that the limiting inverse has the form

$$T^*(u) = \begin{cases} 0, & 0 < u < p, \\ 1, & p < u < 1. \end{cases} \tag{250}$$

Equivalently, the corresponding limiting distribution function is

$$L^*(x) = \begin{cases} 0, & x < 0, \\ p, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases} \tag{251}$$

In particular,

$$L^*(x) = p \qquad \text{for every } x \in (0,1). \tag{252}$$

Let $c_n \in (0,1)$ denote the unique interior fixed point from *Lemma 60*. By passing to a further subsequence if necessary, and using (239), we may assume that

$$c_{n_k} \to c \qquad \text{for some } c \in [\varepsilon, 1-\varepsilon]. \tag{253}$$

By the crossing pattern,

$$L_n(x) < x \quad \text{for } x \in (0, c_n), \qquad L_n(x) > x \quad \text{for } x \in (c_n, 1). \tag{254}$$

We now derive a contradiction.
If $p = 0$, choose $x \in (c, 1)$. This is possible because $c \le 1 - \varepsilon < 1$. Then, for all sufficiently large k, we have $x > c_{n_k}$, and hence

$$L_{n_k}(x) > x.$$

Passing to the limit at the interior continuity point x of L^*, and using (252), gives

$$0 = L^*(x) \ge x,$$

which is impossible.

If $p = 1$, choose $x \in (0, c)$. This is possible because $c \ge \varepsilon > 0$. Then, for all sufficiently large k, we have $x < c_{n_k}$, and hence

$$L_{n_k}(x) < x.$$

Passing to the limit gives

$$1 = L^*(x) \le x,$$

which is impossible.
Finally, suppose $0 < p < 1$. If $p < c$, choose $x \in (p, c)$. Then $x < c_{n_k}$ for all sufficiently large k, so

$$L_{n_k}(x) < x.$$

Passing to the limit gives

$$p = L^*(x) \le x,$$

which contradicts $x < p$. If $p > c$, choose $x \in (c, p)$. Then $x > c_{n_k}$ for all sufficiently large k, so

$$L_{n_k}(x) > x.$$

Passing to the limit gives

$$p = L^*(x) \ge x,$$

which contradicts $x > p$. If $p = c$, choose $x \in (0, c)$. Then $x < c_{n_k}$ for all sufficiently large k, and passing to the limit gives $p \le x$, which is impossible because $x < c = p$.
Thus all possible values of $p \in [0,1]$ lead to contradictions. Therefore our assumption (245) was false. Hence there exists a constant $C > 0$ such that

$$I_n \ge C > 0 \qquad \text{for all } n \ge 0. \tag{255}$$

It follows that

$$0 \le L'_{n+1}(x) = \frac{w(T_n(x))}{I_n} \le \frac{1/4}{C} = M_1 \tag{256}$$

for all $x \in [0,1]$ and all $n \ge 0$.
It remains only to include L_0. From the definition of L_0,

$$L_0(x) = \frac{\int_0^x F^{-1}(u)(1 - F^{-1}(u))\,du}{\int_0^1 F^{-1}(u)(1 - F^{-1}(u))\,du},$$

we have

$$L_0'(x) = \frac{F^{-1}(x)(1 - F^{-1}(x))}{\int_0^1 F^{-1}(u)(1 - F^{-1}(u))\,du}.$$

Since $0 \le F^{-1}(x) \le 1$, the numerator is bounded by $1/4$. Therefore

$$0 \le L_0'(x) \le \frac{1/4}{\int_0^1 F^{-1}(u)(1-F^{-1}(u))\,du} = M_0. \tag{257}$$

Set

$$M = \max\{M_0, M_1\}. \tag{258}$$

Then

$$0 \le L_n'(x) \le M$$

for all $x \in [0,1]$ and all $n \ge 0$.
By the *Mean Value Theorem*, for any $x, y \in [0,1]$ and every $n \ge 0$, there exists ξ between x and y such that

$$|L_n(x) - L_n(y)| = |L_n'(\xi)|\,|x-y| \le M|x-y|. \tag{259}$$

This proves the *uniform Lipschitz bound.* □

Corollary 63. *The family $(L_n)_{n\ge 0}$ is equicontinuous and uniformly bounded on $[0,1]$.*

Proof. Equicontinuity is immediate from the *uniform Lipschitz bound*: given $\varepsilon > 0$, set $\delta = \frac{\varepsilon}{M}$. Then for any n and any x, y with $|x-y| < \delta$ we have

$$|L_n(x) - L_n(y)| \le M|x-y| < \varepsilon. \tag{260}$$

Uniform boundedness follows since each L_n maps $[0,1]$ into $[0,1]$.

We may conclude that the derivative formula $L_{n+1}'(x) = \frac{w(T_n(x))}{I_n}$ shows that the slope of L_{n+1} at any point is a ratio of a local value of $w(T_n)$ to its global average I_n. As w is uniformly bounded and I_n is uniformly separated from zero, the slopes cannot explode. Thus no L_n can develop arbitrarily steep spikes: the whole family is *uniformly Lipschitz* and hence equicontinuous. □

By the *Arzelà–Ascoli's theorem* (see, e.g., [55, Theorem 11.28] or [54, Theorem 14]) we obtain the following compactness result.

Claim 64. *The family $(L_n)_{n\ge 0}$ is relatively compact in $C([0,1])$ with the uniform topology. That is, for every sequence $\{L_{n_k}\}_{n_k=0}^{\infty}$ there exists a subsequence $\{L_{n_{k_j}}\}_{n_{k_j}\in N}$ and a continuous increasing function $L^* : [0,1] \to [0,1]$ such that*

$$\sup_{x\in[0,1]} |L_{n_{k_j}}(x) - L^*(x)| \longrightarrow 0 \quad \text{as } j \to \infty. \tag{261}$$

Moreover, $L^*(0) = 0$ and $L^*(1) = 1$.

Remark 65. At this stage we do *not* claim that the full sequence $\{L_n\}_{n=0}^{\infty}$ converges. We only know that every sequence of indices admits a subsequence along which L_n converges uniformly to some limit L^*, and we will now derive strong structural properties of any such L^*.

Lemma 66. *Let $\{f_k\}_{k=0}^{\infty}$ be a sequence of strictly increasing continuous maps from $[0,1]$ onto $[0,1]$ that converges uniformly to a strictly increasing continuous map $f : [0,1] \to [0,1]$. Then the inverses f_k^{-1} converge uniformly to f^{-1} on $[0,1]$.*

Proof. This is a standard result about strictly monotone homeomorphisms on a compact interval; for completeness we sketch the argument. Fix $\varepsilon > 0$. By uniform continuity of f^{-1} there exists $\delta > 0$ such that $|u-v| < \delta$ implies $|f^{-1}(u) - f^{-1}(v)| < \varepsilon$.

The uniform convergence $f_k \to f$ implies there exists K such that for all $k \ge K$,

$$\sup_{x\in[0,1]} |f_k(x) - f(x)| < \delta. \tag{262}$$

Fix $y \in [0,1]$ and $k \ge K$. Let $x_k = f_k^{-1}(y)$ and $x = f^{-1}(y)$. Then

$$|f(x_k) - f(x)| \le |f(x_k) - f_k(x_k)| + |f_k(x_k) - f(x)| = |f(x_k) - f_k(x_k)| + |y - f(x)| < \delta + 0 = \delta. \tag{263}$$

Hence $|x_k - x| = |f^{-1}(f(x_k)) - f^{-1}(f(x))| < \varepsilon$. Since y and k were arbitrary (for $k \geq K$), this yields uniform convergence $f_k^{-1} \to f^{-1}$ on $[0,1]$. □

We now analyze the locations where L_n crosses the diagonal. For each $n \geq 0$, *Lemma 60* provides a unique point

$$c_n \in (0,1) \tag{264}$$

such that $L_n(c_n) = c_n$ and $L_n(x) < x$ for $x < c_n$, $L_n(x) > x$ for $x > c_n$.

Since $\{c_n\}_{n=0}^{\infty} \subset [0,1]$ is bounded, it admits convergent subsequences. Our next goal is to derive, subsequence by subsequence, a mean-value identity satisfied by any such limit point.

Claim 67. *Let $\{n_k\}_{k=1}^{\infty}$ be a strictly increasing sequence of indices with $n_k \to \infty$. Suppose that*

$$c_{n_k} \longrightarrow c^* \in (0,1) \tag{265}$$

along this subsequence. Then there exists a further subsequence (still denoted n_k) and a strictly increasing continuous map $L^* : [0,1] \to [0,1]$ with inverse $T^* = (L^*)^{-1}$ such that

$$L_{n_k-1} \to L^*, \qquad T_{n_k-1} \to T^* \quad \textit{uniformly on } [0,1],$$

and c^* satisfies the mean-value equation

$$\int_0^{c^*} w(T^*(u))\, du = c^* \int_0^1 w(T^*(u))\, du. \tag{266}$$

Proof. Fix a subsequence $\{n_k\}_{k=0}^{\infty}$ with $c_{n_k} \to c \in (0,1)$. Since $n_k \to \infty$, the shifted indices $m_k = n_k - 1$ also satisfy $m_k \to \infty$.

By *Claim 64*, the family $(L_n)_{n\geq 0}$ is relatively compact in $C([0,1])$. Hence from the sequence $\{L_{m_k}\}_{k=1}^{\infty}$ we can extract a further subsequence (not relabelled) such that

$$L_{m_k} \to L^* \quad \text{uniformly on } [0,1] \tag{267}$$

for some continuous increasing $L^* : [0,1] \to [0,1]$ with $L^*(0) = 0$, $L^*(1) = 1$. By *Lemma 66*, the inverses

$$T_{m_k} = L_{m_k}^{-1} \tag{268}$$

then converge uniformly to $T^* = (L^*)^{-1}$ on $[0,1]$.

For each k we have $L_{n_k}(c_{n_k}) = c_{n_k}$. By the representation (227) applied with $n = m_k = n_k - 1$,

$$L_{n_k}(x) = \frac{\int_0^x w(T_{n_k-1}(u))\, du}{\int_0^1 w(T_{n_k-1}(u))\, du} = \frac{\int_0^x w(T_{m_k}(u))\, du}{\int_0^1 w(T_{m_k}(u))\, du}, \tag{269}$$

where $T_{m_k} = L_{m_k}^{-1}$. Evaluating at $x = c_{n_k}$ and using $L_{n_k}(c_{n_k}) = c_{n_k}$ gives

$$c_{n_k} = \frac{\int_0^{c_{n_k}} w(T_{m_k}(u))\, du}{\int_0^1 w(T_{m_k}(u))\, du}. \tag{270}$$

Equivalently,

$$\int_0^{c_{n_k}} w(T_{m_k}(u))\, du = c_{n_k} \int_0^1 w(T_{m_k}(u))\, du. \tag{271}$$

By uniform convergence $T_{m_k} \to T^*$ and continuity of w, we have uniform convergence

$$w \circ T_{m_k} \to w \circ T^* \quad \text{on } [0,1].$$

Thus, for any fixed $x \in [0,1]$,

$$\int_0^x w(T_{m_k}(u))\, du \longrightarrow \int_0^x w(T^*(u))\, du, \quad \int_0^1 w(T_{m_k}(u))\, du \longrightarrow \int_0^1 w(T^*(u))\, du. \tag{272}$$

Moreover, $c_{n_k} \to c$ by assumption.

We now pass to the limit in (270). For the left-hand side we use continuity of

$$x \mapsto \int_0^x w(T^*(u))\,du \tag{273}$$

and the convergence $c_{n_k} \to c^*$ together with (272). For the right-hand side, combine $c_{n_k} \to c$ with (272). The limit of (271) as $k \to \infty$ is therefore

$$\int_0^{c^*} w(T^*(u))\,du = c^* \int_0^1 w(T^*(u))\,du, \tag{274}$$

which is exactly (266). □

Remark 68. Claim 67 is purely subsequential: we fix one convergent subsequence $\{c_{n_k}\}_{k=1}^{\infty}$, and from it we extract a companion subsequence of $\{L_{n_k-1}\}_{k=1}^{\infty}$ with limit L^*. There is no claim that different subsequences of $\{c_n\}_{n=0}^{\infty}$ must share the same limit c^*, nor that different subsequential limits L^* coincide.

To understand the mean-value equation (266), we analyze the shape of the weight function

$$w^*(u) = w(T^*(u)), \qquad u \in [0,1]. \tag{275}$$

Lemma 69. *Let L^* and $T^* = (L^*)^{-1}$ be as in Lemma 67, and define*

$$w^*(u) = w(T^*(u)) = T^*(u)(1 - T^*(u)). \tag{276}$$

Then:
1. *w^* is continuous on $[0,1]$, satisfies $w^*(0) = w^*(1) = 0$, and is strictly positive on $(0,1)$;*
2. *there exists a unique $u_0 \in (0,1)$ such that $T^*(u_0) = \frac{1}{2}$ and*

$$w^* \text{ is strictly increasing on } [0,u_0], \quad w^* \text{ is strictly decreasing on } [u_0,1]. \tag{277}$$

In particular, w^* is strictly unimodal on $[0,1]$ with a single peak at u_0.

Proof. Since L^* is continuous, strictly increasing, and maps $[0,1]$ onto $[0,1]$, its inverse T^* is also continuous, strictly increasing, with $T^*(0) = 0$ and $T^*(1) = 1$. The function $w(y) = y(1-y)$ is continuous on $[0,1]$, equals 0 at $y = 0$ and $y = 1$, and is strictly positive on $(0,1)$.

Thus $w^* = w \circ T^*$ is continuous on $[0,1]$, satisfies

$$w^*(0) = w(0) = 0, \quad w^*(1) = w(1) = 0, \tag{278}$$

and is strictly positive on $(0,1)$ because $T^*(u) \in (0,1)$ for $u \in (0,1)$ and $w(y) > 0$ for $y \in (0,1)$. This proves (1).

For (2), note that T^* is strictly increasing and onto $[0,1]$, so there exists a unique $u_0 \in (0,1)$ such that $T^*(u_0) = \frac{1}{2}$. Since T^* is strictly increasing, we have

$$T^*(u) < \tfrac{1}{2} \quad \text{for } u < u_0, \qquad T^*(u) > \tfrac{1}{2} \quad \text{for } u > u_0. \tag{279}$$

The kernel w is strictly increasing on $[0, \frac{1}{2}]$ and strictly decreasing on $[\frac{1}{2}, 1]$. Therefore, for $u_1 < u_2 \le u_0$ we have

$$T^*(u_1) < T^*(u_2) \le \tfrac{1}{2} \implies w(T^*(u_1)) < w(T^*(u_2)), \tag{280}$$

so w^* is strictly increasing on $[0,u_0]$. Similarly, for $u_0 \le u_1 < u_2$ we have

$$\tfrac{1}{2} \le T^*(u_1) < T^*(u_2) \implies w(T^*(u_1)) > w(T^*(u_2)), \tag{281}$$

so w^* is strictly decreasing on $[u_0,1]$. Thus w^* is strictly unimodal with a unique maximum at u_0. □

With this shape information we can now analyze the mean-value equation (274) for a general strictly unimodal weight.

Claim 70. *Let $f : [0,1] \to [0,\infty)$ be continuous, strictly positive on $(0,1)$, and strictly unimodal in the sense that there exists $u_0 \in (0,1)$ such that f is strictly increasing on $[0,u_0]$ and strictly decreasing on $[u_0,1]$. Define*

$$F(c) = \int_0^c f(u)\,du - c\int_0^1 f(u)\,du, \qquad 0 \le c \le 1. \tag{282}$$

Then the equation

$$\int_0^c f(u)\,du = c\int_0^1 f(u)\,du \tag{283}$$

has at most one solution $c \in (0,1)$.

Proof. The function F defined in (282) is continuously differentiable, with

$$F(0) = 0, \quad F(1) = 0, \qquad F'(c) = f(c) - \int_0^1 f(u)\,du = f(c) - C, \tag{284}$$

where $C = \int_0^1 f(u)\,du > 0$. Thus (283) is equivalent to $F(c) = 0$.

By strict unimodality, the graph of f increases up to a single maximum and then decreases. Therefore the horizontal line $y = C$ can intersect the graph of f in at most two points in $(0,1)$. In other words, the equation

$$f(c) = C \tag{285}$$

has at most two solutions in $(0,1)$. Equivalently, $F'(c) = 0$ has at most two zeros in $(0,1)$.

Suppose, by contradiction, that $F(c) = 0$ has two distinct solutions in $(0,1)$, say $0 < c_1 < c_2 < 1$, in addition to $c = 0$ and $c = 1$. By *Rolle's Theorem*, there exist points

$$c_1' \in (0,c_1), \quad c_2' \in (c_1,c_2), \quad c_3' \in (c_2,1) \tag{286}$$

such that

$$F'(c_1') = F'(c_2') = F'(c_3') = 0. \tag{287}$$

Thus F' would have at least three distinct zeros in $(0,1)$, contradicting the fact that F' can have at most two zeros. Hence $F(c) = 0$ has at most one solution in $(0,1)$. □

We can now combine *Claims 67, 69, and 70* to characterise the subsequential limits of $\{c_n\}_{n=0}^{\infty}$.

Corollary 71. *Let $\{n_k\}_{k=0}^{\infty}$ be a strictly increasing sequence with $n_k \to \infty$, such that $c_{n_k} \to c^* \in (0,1)$. Then there exists a subsequential limit T^* of the inverses $\{T_{n_k-1}\}_{k=0}^{\infty}$ such that c^* is the unique solution in $(0,1)$ of*

$$\int_0^{c^*} w(T^*(u))\,du = c^*\int_0^1 w(T^*(u))\,du. \tag{288}$$

In particular, for this fixed subsequence and its associated limit T^*, the interior crossing points c_{n_k} cannot converge to two different values.

Proof. By *Claim 67*, there exists a further subsequence of $\{n_k\}_{k=0}^{\infty}$ and a limit T^* such that (274) holds. By *Lemma 69*, the function

$$f(u) = w(T^*(u)) \tag{289}$$

is continuous, strictly positive on $(0,1)$, and strictly unimodal. Therefore *Claim 70* applies and shows that the mean-value equation (288) has at most one solution $c^* \in (0,1)$.

Since $c_{n_k} \to c^*$ and (274) holds with this c, we see that c^* is indeed a solution of (288); by uniqueness, no other limit in $(0,1)$ is possible for this subsequence and this T^*. □

Remark 72. *Corollary 71* should be read carefully. It says: Fix a convergent subsequence $\{c_{n_k}\}_{k=0}^{\infty}$ with limit c^*. Extract from $\{n_k\}_{k\geq 0}$ a further subsequence along which T_{n_k-1} converges to some T^*. For this particular T^*, the mean-value equation (288) has a unique interior solution, and the limit of $\{c_{n_k}\}_{k=0}^{\infty}$ must be that solution.

What it does not claim is that:
- different subsequences $\{n_k\}_{k\geq 0}$ and $\{m_\ell\}_{l\geq 0}$ produce the same limit T^*; or
- the corresponding mean-value points c^* must coincide across all subsequences.

Corollary 73. *It holds*

$$L^*(x) < x \ (x < c^*), \qquad L^*(x) > x \ (x > c^*).$$

as well as

$$T^*(x) > x \ (x < c^*), \qquad T^*(x) < x \ (x > c^*).$$

Proof. Fix $\varepsilon > 0$ small. For $x \leq c^* - \varepsilon$ and k large we have $x < c_{n_k}$, hence by *Lemma 60* we have $L_{n_k}(x) < x$; passing to the limit gives $L^*(x) \leq x$. Similarly, for $x \geq c^* + \varepsilon$ and k large we have $x > c_{n_k}$ and thus $L_{n_k}(x) > x$, so $L^*(x) \geq x$. Letting $\varepsilon \downarrow 0$ and using continuity of L^* yields $L^*(x) < x$ for $x < c^*$ and $L^*(x) > x$ for $x > c^*$. □

Remark 74. At this stage of the argument, it is still logically possible that different subsequences of $\{c_n\}_{n=0}^{\infty}$ converge to different limits $c^{*(1)}, c^{*(2)}, \ldots$, each associated with its own subsequential limit $T^{*(i)}, i = 1, 2, ...$ and its own mean-value equation. The task of the subsequent analysis is precisely to exclude this possibility and show that all such constants must in fact coincide.

Symmetry defect of L_n

In this section, we characterize the symmetry of L_n with respect to $\frac{1}{2}$. We begin with several general observations before proceeding to more complex derivations.

Lemma 75. *Let $L : [0,1] \to [0,1]$ be a continuous strictly increasing map with inverse $T = L^{-1}$. Define the symmetry defects*

$$D_L(x) = L(x) + L(1-x) - 1, \qquad x \in [0,1], \tag{290}$$

and

$$D_T(u) = T(u) + T(1-u) - 1, \qquad u \in [0,1]. \tag{291}$$

Then the following implications hold

$$D_L(x) \leq 0 \ \forall x \in [0,1] \quad \Longrightarrow \quad D_T(u) \geq 0 \ \forall u \in [0,1], \tag{292}$$

and

$$D_L(x) \geq 0 \ \forall x \in [0,1] \quad \Longrightarrow \quad D_T(u) \leq 0 \ \forall u \in [0,1]. \tag{293}$$

In particular, whenever D_L has a constant sign on $[0,1]$, D_T has the opposite constant sign on $[0,1]$.

Proof. Assume first that

$$L(x) + L(1-x) \leq 1 \qquad \forall x \in [0,1]. \tag{294}$$

Fix $u \in [0,1]$ and set $x = T(u)$, so that $u = L(x)$. Then (294) gives

$$L(1-x) \leq 1 - L(x) = 1 - u. \tag{295}$$

Since T is increasing,

$$1 - x = T(L(1-x)) \leq T(1-u). \tag{296}$$

Therefore

$$T(u) + T(1-u) = x + T(1-u) \geq x + (1-x) = 1, \tag{297}$$

that is,

$$D_T(u) = T(u) + T(1-u) - 1 \geq 0. \tag{298}$$

This proves (292). The proof of (293) is identical, with all inequalities reversed. □

Going back to the usual indexed notation we have the following:

Lemma 76. *Assume that the initial defect*

$$D_0(x) = L_0(x) + L_0(1-x) - 1 \tag{299}$$

has a constant sign on $[0,1]$. *Then, for every* $n \geq 0$, *the defect*

$$D_n(x) = L_n(x) + L_n(1-x) - 1, \qquad x \in [0,1], \tag{300}$$

has a constant sign on $[0,1]$, *the sign alternates with* n, *and* $|D_n|$ *is nondecreasing on* $[0, \frac{1}{2}]$.

Proof. For brevity write

$$S_n(x) = T_n(x) + T_n(1-x) - 1, \qquad x \in [0,1], \tag{301}$$

where again $T_n = L_n^{-1}$. By *Lemma 75*, S_n has the opposite constant sign to D_n.

Differentiating

$$D_{n+1}(x) = L_{n+1}(x) + L_{n+1}(1-x) - 1 \tag{302}$$

and using

$$L'_{n+1}(x) = \frac{w(T_n(x))}{\int_0^1 w(T_n(u))\,du}, \qquad w(y) = y(1-y), \tag{303}$$

we obtain

$$\begin{aligned} D'_{n+1}(x) &= L'_{n+1}(x) - L'_{n+1}(1-x) \\ &= \frac{w(T_n(x)) - w(T_n(1-x))}{\int_0^1 w(T_n(u))\,du}. \end{aligned} \tag{304}$$

Since

$$w(a) - w(b) = (a-b)(1-a-b), \tag{305}$$

we get

$$D'_{n+1}(x) = \frac{(T_n(1-x) - T_n(x))(T_n(x) + T_n(1-x) - 1)}{\int_0^1 w(T_n(u))\,du}. \tag{306}$$

Hence, for $x \in [0, \frac{1}{2}]$, because T_n is increasing,

$$T_n(1-x) - T_n(x) \geq 0, \tag{307}$$

and therefore

$$\operatorname{sgn}(D'_{n+1}(x)) = \operatorname{sgn}(S_n(x)), \qquad x \in [0, \tfrac{1}{2}]. \tag{308}$$

Now assume inductively that D_n has a constant sign on $[0,1]$. Then S_n has the opposite constant sign on $[0,1]$, so by (308) the derivative D'_{n+1} has that opposite sign on $[0, \frac{1}{2}]$. Since

$$D_{n+1}(0) = L_{n+1}(0) + L_{n+1}(1) - 1 = 0, \tag{309}$$

for every $x \in [0, \frac{1}{2}]$ we have

$$D_{n+1}(x) = D_{n+1}(0) + \int_0^x D'_{n+1}(t)\,dt = \int_0^x D'_{n+1}(t)\,dt. \tag{310}$$

Thus, if D'_{n+1} has a constant sign on $[0, \frac{1}{2}]$, then D_{n+1} has the same constant sign on $[0, \frac{1}{2}]$. By symmetry,

$$D_{n+1}(1-x) = D_{n+1}(x), \qquad x \in [0,1], \tag{311}$$

hence D_{n+1} has that constant sign on all of $[0,1]$. This proves that the sign alternates.

Moreover, on $[0,\frac{1}{2}]$, the sign-corrected function

$$\varepsilon_n\, D_n(x), \qquad \varepsilon_n = -\operatorname{sgn}(D_n(x)) \in \{-1,+1\}, \tag{312}$$

has nonnegative derivative by (308). Hence $\varepsilon_n D_n$ is nondecreasing on $[0,\frac{1}{2}]$, i.e.

$$|D_n(x)| \text{ is nondecreasing on } [0,\tfrac{1}{2}]. \tag{313}$$

The induction is complete. ☐

Claim 77. *Assume the hypotheses of Lemma 76. Define*

$$A_n = \int_0^1 |D_n(x)|\,dx, \tag{314}$$

and let

$$E_n = \int_0^1 (1 - L_n(x))\,dx \tag{315}$$

be the expectation of the random variable with distribution function L_n. Then

$$A_n = |1 - 2E_n| \tag{316}$$

for every n, and the sequence $\{A_n\}_{n\geq 0}$ is nonincreasing:

$$A_{n+1} \leq A_n \qquad \textit{for all } n \geq 0. \tag{317}$$

Proof. Since D_n has a constant sign on $[0,1]$ by *Lemma 76*, we have

$$A_n = \left| \int_0^1 D_n(x)\,dx \right|. \tag{318}$$

Also,

$$\begin{aligned}\int_0^1 D_n(x)\,dx &= \int_0^1 (L_n(x) + L_n(1-x) - 1)\,dx \\ &= 2\int_0^1 L_n(x)\,dx - 1 \\ &= 1 - 2E_n, \end{aligned} \tag{319}$$

because

$$E_n = \int_0^1 (1 - L_n(x))\,dx. \tag{320}$$

This proves (316).

We now prove (317). For brevity write

$$L = L_n, \qquad T = T_n, \qquad g(x) = 1 - L(x), \qquad E = E_n. \tag{321}$$

Also set as before

$$I = \int_0^1 w(T(u))\,du. \tag{322}$$

From the iterative equation defining L_{n+1},

$$L_{n+1}(x) = \frac{\int_0^x w(T(u))\,du}{I}, \tag{323}$$

we have

$$\begin{aligned} E_{n+1} &= \int_0^1 (1 - L_{n+1}(x))\,dx = 1 - \int_0^1 L_{n+1}(x)\,dx \\ &= 1 - \frac{1}{I}\int_0^1 \left(\int_0^x w(T(u))\,du\right) dx \\ &= 1 - \frac{1}{I}\int_0^1 \left(\int_u^1 dx\right) w(T(u))\,du \\ &= 1 - \frac{1}{I}\int_0^1 (1-u)w(T(u))\,du. \end{aligned} \tag{324}$$

Therefore

$$I\,(E + E_{n+1} - 1) = IE - \int_0^1 (1-u)w(T(u))\,du. \tag{325}$$

We next rewrite I and the numerator in (324) in terms of g. Using the change of variables $u = L(x)$, so that $du = L'(x)\,dx$ and $T(u) = x$, we obtain

$$I = \int_0^1 x(1-x)L'(x)\,dx = \int_0^1 (1-2x)g(x)\,dx. \tag{326}$$

Similarly,

$$\begin{aligned} \int_0^1 (1-u)w(T(u))\,du &= \int_0^1 g(x)\,x(1-x)L'(x)\,dx \\ &= \frac{1}{2}\int_0^1 (1-2x)g(x)^2\,dx. \end{aligned} \tag{327}$$

Now define on $[0, \frac{1}{2}]$

$$r(x) = g(x) + g(1-x) - 1 = -D_n(x), \tag{328}$$

$$d(x) = g(x) - g(1-x), \tag{329}$$

and

$$p(x) = (1-2x)\,d(x). \tag{330}$$

Since g is decreasing, $d(x) \geq 0$ on $[0, \frac{1}{2}]$. Moreover, if $0 \leq x \leq y \leq \frac{1}{2}$, then

$$d(x) - d(y) = (g(x) - g(y)) + (g(1-y) - g(1-x)) \geq 0, \tag{331}$$

so d is nonincreasing on $[0, \frac{1}{2}]$. Therefore p is nonnegative and nonincreasing on $[0, \frac{1}{2}]$.

Pairing x with $1-x$ in (326) gives

$$I = \int_0^{1/2} p(x)\,dx. \tag{332}$$

Likewise, pairing x with $1-x$ in (327) yields

$$\int_0^1 (1-u)w(T(u))\,du = \frac{1}{2}\int_0^{1/2} (1-2x)(g(x)^2 - g(1-x)^2)\,dx = \frac{1}{2}\int_0^{1/2} p(x)(1+r(x))\,dx. \tag{333}$$

Furthermore,

$$E - \frac{1}{2} = \int_0^{1/2} r(x)\,dx. \tag{334}$$

We now substitute (332), (333) and the identity

$$E = \frac{1}{2} + \int_0^{1/2} r(x)\,dx \tag{335}$$

into (325). This gives

$$I\,(E + E_{n+1} - 1) = I\left(\frac{1}{2} + \int_0^{1/2} r(x)\,dx\right) - \frac{1}{2}\int_0^{1/2} p(x)(1+r(x))\,dx = \frac{1}{2}I + I\int_0^{1/2} r(x)\,dx - \frac{1}{2}\int_0^{1/2} p(x)\,dx - \frac{1}{2}\int_0^{1/2} p(x)r(x)\,dx. \tag{336}$$

Using (332), the two scalar terms cancel

$$\frac{1}{2}I - \frac{1}{2}\int_0^{1/2} p(x)\,dx = 0. \tag{337}$$

Hence

$$I\,(E + E_{n+1} - 1) = \left(\int_0^{1/2} r(x)\,dx\right)\left(\int_0^{1/2} p(x)\,dx\right) - \frac{1}{2}\int_0^{1/2} p(x)r(x)\,dx. \tag{338}$$

Let

$$\sigma = \operatorname{sgn}\left(E - \frac{1}{2}\right). \tag{339}$$

Since $r = -D_n$ and D_n has a constant sign, r also has a constant sign on $[0, \frac{1}{2}]$. Moreover,

$$E - \frac{1}{2} = \int_0^{1/2} r(x)\,dx. \tag{340}$$

Hence, if $E \neq \frac{1}{2}$, the sign of r coincides with

$$\sigma = \operatorname{sgn}\left(E - \frac{1}{2}\right), \tag{341}$$

and therefore

$$\sigma r(x) = |r(x)| \qquad \text{for all } x \in [0, \tfrac{1}{2}]. \tag{342}$$

If $E = \frac{1}{2}$, then $\int_0^{1/2} r(x)\,dx = 0$; since r has a constant sign, this implies $r \equiv 0$, and the argument is trivial. Thus the function

$$q(x) = \sigma r(x) = |r(x)| \tag{343}$$

is nonnegative on $[0, \frac{1}{2}]$. By *Lemma 76*, $|D_n|$ is nondecreasing on $[0, \frac{1}{2}]$, hence $q = |r| = |D_n|$ is also nondecreasing on $[0, \frac{1}{2}]$.

Multiplying (338) by σ we obtain

$$I\,\sigma(E + E_{n+1} - 1) = \left(\int_0^{1/2} q(x)\,dx\right)\left(\int_0^{1/2} p(x)\,dx\right) - \frac{1}{2}\int_0^{1/2} p(x)q(x)\,dx. \tag{344}$$

Now p is nonnegative and nonincreasing on $[0, \frac{1}{2}]$, whereas q is nonnegative and nondecreasing on $[0, \frac{1}{2}]$. Hence the *reverse Chebyshev inequality* (see [26] and [36, Chapter 9]) on the interval $[0, \frac{1}{2}]$ yields

$$\int_0^{1/2} p(x)q(x)\,dx \le 2\left(\int_0^{1/2} p(x)\,dx\right)\left(\int_0^{1/2} q(x)\,dx\right). \tag{345}$$

Substituting (345) into (344) gives

$$I\,\sigma(E + E_{n+1} - 1) \ge 0. \tag{346}$$

Since $I > 0$, we conclude that

$$\left(E_n - \frac{1}{2}\right)(E_n + E_{n+1} - 1) \ge 0. \tag{347}$$

Finally, by *Lemma 76*, the sign of D_n alternates with n, hence the sign of $r = -D_n$, and therefore also the sign of $E_n - \frac{1}{2}$, alternates with n. Thus

$$|E_n - \tfrac{1}{2}| - |E_{n+1} - \tfrac{1}{2}| = \sigma\,(E_n + E_{n+1} - 1) \ge 0. \tag{348}$$

Therefore

$$|E_{n+1} - \tfrac{1}{2}| \le |E_n - \tfrac{1}{2}|, \tag{349}$$

and by (318),

$$A_{n+1} = 2|E_{n+1} - \tfrac{1}{2}| \le 2|E_n - \tfrac{1}{2}| = A_n. \tag{350}$$

This proves (317). ☐

Corollary 78. *Assume the hypotheses of Claim 77. Fix $n \ge 0$ and define, on $[0, \frac{1}{2}]$,*

$$p_n(x) = (1 - 2x)(L_n(1 - x) - L_n(x)), \tag{351}$$

and

$$q_n(x) = |D_n(x)| = |L_n(x) + L_n(1 - x) - 1|. \tag{352}$$

If
1. $A_n > 0$,
2. p_n *is strictly decreasing on* $[0, \frac{1}{2}]$,
3. q_n *is strictly increasing on* $[0, \frac{1}{2}]$,

then

$$\left(E_n - \frac{1}{2}\right)(E_n + E_{n+1} - 1) > 0, \tag{353}$$

and consequently

$$A_{n+1} < A_n. \tag{354}$$

In particular, if the above three conditions hold for every index n with $A_n > 0$, then $\{A_n\}_{n\ge0}$ is strictly decreasing until it reaches 0.

Proof. We use the notation introduced in the proof of *Claim 77*. There we obtained

$$I\,\sigma(E_n+E_{n+1}-1)=\left(\int\limits_0^{1/2} q_n(x)\,dx\right)\left(\int\limits_0^{1/2} p_n(x)\,dx\right)-\frac{1}{2}\int\limits_0^{1/2} p_n(x)q_n(x)\,dx, \tag{355}$$

where

$$I=\int\limits_0^1 w(T_n(u))\,du>0, \qquad \sigma=\operatorname{sgn}\Big(E_n-\frac{1}{2}\Big). \tag{356}$$

By assumption, p_n is nonnegative and strictly decreasing on $[0,\frac{1}{2}]$, while q_n is nonnegative and strictly increasing on $[0,\frac{1}{2}]$. Moreover, $A_n>0$ implies

$$\int\limits_0^{1/2} q_n(x)\,dx>0, \tag{357}$$

so q_n is not identically zero.

Hence the *reverse Chebyshev inequality* is strict

$$\int\limits_0^{1/2} p_n(x)q_n(x)\,dx<2\left(\int\limits_0^{1/2} p_n(x)\,dx\right)\left(\int\limits_0^{1/2} q_n(x)\,dx\right). \tag{358}$$

Substituting (358) into (355) yields

$$I\,\sigma(E_n+E_{n+1}-1)>0. \tag{359}$$

Since $I>0$, we conclude that

$$\Big(E_n-\frac{1}{2}\Big)(E_n+E_{n+1}-1)>0. \tag{360}$$

Finally, as in the proof of *Claim 77*, the sign of $E_n-\frac{1}{2}$ alternates with n, and

$$A_n=2\left|E_n-\frac{1}{2}\right|. \tag{361}$$

Therefore (360) is equivalent to

$$\left|E_{n+1}-\frac{1}{2}\right|<\left|E_n-\frac{1}{2}\right|, \tag{362}$$

which in turn gives

$$A_{n+1}<A_n. \tag{363}$$

This proves the claim. □

Remark 79. The additional assumptions in *Corollary 78* are natural from the point of view of the geometry of the defect. Indeed, on $[0,\frac{1}{2}]$ the quantity

$$q_n(x)=|D_n(x)|=|L_n(x)+L_n(1-x)-1| \tag{364}$$

measures the magnitude of the defect, and the inductive sign pattern proved in *Lemma 76* shows that q_n is nondecreasing on $[0,\frac{1}{2}]$. Thus the additional strict monotonicity assumption on q_n simply excludes the degenerate situation in which the defect remains locally flat on a nontrivial subinterval.

Likewise,

$$p_n(x)=(1-2x)(L_n(1-x)-L_n(x)), \qquad x\in[0,\tfrac{1}{2}], \tag{365}$$

is the geometric weight that appears after symmetrizing the identities for I_n and E_{n+1} on the half-interval. The factor $1-2x$ decreases strictly from 1 to 0, while $L_n(1-x)-L_n(x)\geq 0$ measures the separation between the two symmetric branches of L_n. Hence the condition that p_n be strictly decreasing means that this weighted branch-separation does not contain a plateau compensating the decay of $1-2x$.

Under these two strict half-interval shape conditions, the reverse *Chebyshev inequality* used in the proof *Claim 77* becomes strict, and therefore the decay of A_n is strict whenever $A_n > 0$. Numerically, these hypotheses are precisely what is observed in the examples considered in this appendix: the defect profile q_n grows strictly from the endpoint 0 to its extremal value at $x = \frac{1}{2}$, whereas the weighted branch-separation p_n decays strictly to 0 as $x \uparrow \frac{1}{2}$.

Vanishing of the symmetry defect of L_n in the one-sign regime

We now demonstrate the convergence of the symmetry defect to zero, given certain assumptions to be detailed in the next section. This convergence ensures the symmetry of the subsequential limits about $\frac{1}{2}$.

Claim 80. *Assume that there exists an index $N \geq 0$ such that, for every $n \geq N$, the defect*

$$D_n(x) = L_n(x) + L_n(1-x) - 1 \tag{366}$$

has a constant sign on $[0,1]$. *Then*

$$A_n = \int_0^1 |D_n(x)|\,dx \longrightarrow 0 \qquad \text{as } n \to \infty. \tag{367}$$

Proof. Fix $n \geq N$, and define as before on $[0, \frac{1}{2}]$

$$q_n(x) = |D_n(x)|, \tag{368}$$

$$d_n(x) = L_n(1-x) - L_n(x), \tag{369}$$

and

$$p_n(x) = (1-2x)\,d_n(x). \tag{370}$$

Since D_n has a constant sign for every $n \geq N$, the same argument as in *Lemma 76* shows that $q_n = |D_n|$ is nonnegative and nondecreasing on $[0, \frac{1}{2}]$ for every $n \geq N$. On the other hand, the proof of *Claim 77* shows that d_n is nonnegative and nonincreasing on $[0, \frac{1}{2}]$, hence so is p_n.

Moreover,

$$A_n = 2\int_0^{1/2} q_n(x)\,dx. \tag{371}$$

We now extract from the proof of *Claim 77* an exact representation of A_{n+1} as a weighted average of q_n. Indeed, for $n \geq N$, equations (355) and (316) give

$$I_n\,\sigma_n(E_n + E_{n+1} - 1) = \left(\int_0^{1/2} q_n(x)\,dx\right)\left(\int_0^{1/2} p_n(x)\,dx\right) - \frac{1}{2}\int_0^{1/2} p_n(x)q_n(x)\,dx, \tag{372}$$

where

$$I_n = \int_0^1 w(T_n(u))\,du = \int_0^{1/2} p_n(x)\,dx > 0 \tag{373}$$

and

$$\sigma_n = \operatorname{sgn}\Big(E_n - \frac{1}{2}\Big). \tag{374}$$

Since D_n has a constant sign and the sign alternates with n,

$$\sigma_n(E_n + E_{n+1} - 1) = \Big|E_n - \frac{1}{2}\Big| - \Big|E_{n+1} - \frac{1}{2}\Big| = \frac{A_n - A_{n+1}}{2}. \tag{375}$$

Using also

$$\int_0^{1/2} q_n(x)\,dx = \frac{A_n}{2}, \tag{376}$$

$$A_{n+1} = \frac{\int_0^{1/2} p_n(x)q_n(x)\,dx}{\int_0^{1/2} p_n(x)\,dx}, \qquad n \geq N. \tag{377}$$

Next, introduce the two probability measures on $[0, \frac{1}{2}]$

$$d\nu(x) = 2\,dx, \tag{378}$$

and

$$d\lambda(x) = 4(1-2x)\,dx. \tag{379}$$

Then (371) becomes

$$A_n = \int_0^{1/2} q_n\,d\nu. \tag{380}$$

We claim that

$$A_{n+1} \leq \int_0^{1/2} q_n\,d\lambda \leq A_n, \qquad n \geq N. \tag{381}$$

For the first inequality, apply the same *reverse Chebyshev* argument as in the proof of *Claim 77* on $[0, \frac{1}{2}]$ to the nonincreasing function d_n and the nondecreasing function q_n with respect to the base measure $(1-2x)\,dx$ to get

$$\frac{\int_0^{1/2}(1-2x)d_n(x)q_n(x)\,dx}{\int_0^{1/2}(1-2x)d_n(x)\,dx} \leq \frac{\int_0^{1/2}(1-2x)q_n(x)\,dx}{\int_0^{1/2}(1-2x)\,dx}. \tag{382}$$

By (370) and (377), this is exactly

$$A_{n+1} \leq 4\int_0^{1/2}(1-2x)q_n(x)\,dx = \int_0^{1/2} q_n\,d\lambda. \tag{383}$$

For the second inequality, observe that the distribution functions of ν and λ are

$$F_\nu(x) = 2x, \qquad F_\lambda(x) = 4x - 4x^2, \qquad 0 \leq x \leq \frac{1}{2}, \tag{384}$$

and

$$F_\lambda(x) - F_\nu(x) = 2x(1-2x) \geq 0. \tag{385}$$

Thus λ is stochastically smaller than ν, and because q_n is nondecreasing,

$$\int_0^{1/2} q_n\,d\lambda \leq \int_0^{1/2} q_n\,d\nu = A_n. \tag{386}$$

This proves (381).

By the same argument as in *Claim 77*, applied from the index N onward, the tail $\{A_n\}_{n\geq N}$ is nonincreasing. Hence it converges:

$$A_n \downarrow a \qquad \text{for some } a \geq 0.$$

By (381),

$$\int_0^{1/2} q_n\,d\lambda \longrightarrow a. \tag{387}$$

Now the sequence $\{q_n\}_{n\geq N}$ consists of bounded nondecreasing functions on $[0, \frac{1}{2}]$, with $0 \leq q_n \leq 1$. By *Helly's selection principle* there exists a subsequence $\{q_{n_k}\}_{k\geq 1}$ and a bounded nondecreasing function q on $[0, \frac{1}{2}]$ such that

$$q_{n_k}(x) \longrightarrow q(x)$$

for every continuity point of q, hence almost everywhere on $[0, \frac{1}{2}]$. By dominated convergence applied to (380) and (387),

$$\int_0^{1/2} q\, d\nu = a, \qquad \int_0^{1/2} q\, d\lambda = a. \tag{388}$$

Therefore

$$\int_0^{1/2} q\, d\nu - \int_0^{1/2} q\, d\lambda = 0. \tag{389}$$

Since q is nondecreasing, its distributional derivative dq is a nonnegative *Stieltjes measure*. Integration by parts for monotone functions yields

$$\int_0^{1/2} q\, d\nu - \int_0^{1/2} q\, d\lambda = \int_{[0,1/2]} (F_\lambda(x) - F_\nu(x))\, dq(x). \tag{390}$$

But

$$F_\lambda(x) - F_\nu(x) = 2x(1-2x) > 0 \qquad \text{for } 0 < x < \frac{1}{2}, \tag{391}$$

and $dq \geq 0$, so (389) implies

$$dq = 0 \qquad \text{on } (0, \tfrac{1}{2}). \tag{392}$$

Hence q is constant on $[0, \frac{1}{2}]$. Since

$$q_n(0) = |D_n(0)| = 0 \qquad \text{for every } n, \tag{393}$$

we have $q(0) = 0$, and therefore

$$q \equiv 0. \tag{394}$$

Thus (388) gives $a = 0$.

We have shown that every convergent subsequence of $\{A_n\}_{n>N}$ has limit 0. Since $\{A_n\}_{n>N}$ itself converges, its limit must be 0, i.e.

$$A_n \longrightarrow 0. \tag{395}$$

This proves (367). □

Corollary 81. *Assume the hypotheses of Claim 80. Let $\{n_k\}_{k\geq 1}$ be any sequence of integers with $n_k \to \infty$, and suppose that*

$$L_{n_k}(x) \longrightarrow L^*(x) \tag{396}$$

uniformly on $[0,1]$. Then

$$L^*(x) + L^*(1-x) - 1 = 0 \qquad \textit{for every } x \in [0,1]. \tag{397}$$

In particular, under any of the sufficient hypotheses on the starting law F given earlier in this appendix for which the one-sign regime holds from the start, Claim 80 applies with $N = 0$. Hence, by the compactness result Claim 64, every subsequential limit of $\{L_n\}_{n\geq 0}$ in $C([0,1])$ is symmetric.

Proof. By *Claim 80*,

$$A_n = \int_0^1 |D_n(x)|\, dx \longrightarrow 0. \tag{398}$$

Along the subsequence (n_k), define

$$D_{n_k}(x) = L_{n_k}(x) + L_{n_k}(1-x) - 1. \tag{399}$$

Since $L_{n_k} \to L^*$ uniformly on $[0,1]$, we also have

$$D_{n_k} \longrightarrow D^* \tag{400}$$

uniformly on $[0,1]$, where

$$D^*(x) = L^*(x) + L^*(1-x) - 1. \tag{401}$$

Therefore

$$\int_0^1 |D^*(x)|\,dx = \lim_{k\to\infty} \int_0^1 |D_{n_k}(x)|\,dx = \lim_{k\to\infty} A_{n_k} = 0. \tag{402}$$

Since D^* is continuous on $[0,1]$, it follows that

$$D^*(x) = 0 \qquad \forall x \in [0,1]. \tag{403}$$

This proves (397). □

Corollary 82. *Assume the hypotheses of Claim 80. Assume moreover the earlier one-crossing framework: for each n, L_n has a unique crossing point $c_n \in (0,1)$ with the diagonal,*

$$L_n(c_n) = c_n, \tag{404}$$

Then every cluster point of (c_n) is equal to $\frac{1}{2}$. Consequently,

$$c_n \longrightarrow \frac{1}{2}. \tag{405}$$

Proof. Let c^* be any cluster point of $\{c_n\}_{n\geq 0}$. Then there exists a subsequence $\{n_k\}_{k\geq 0}$ such that

$$c_{n_k} \to c^*. \tag{406}$$

By the compactness result *Claim 64*, after passing to a further subsequence if necessary, we may assume that

$$L_{n_k} \to L^* \tag{407}$$

uniformly on $[0,1]$, for some continuous increasing function L^* with $L^*(0) = 0$ and $L^*(1) = 1$. By *Corollary 81*, L^* is symmetric:

$$L^*(x) + L^*(1-x) - 1 = 0 \qquad \text{for every } x \in [0,1]. \tag{408}$$

Moreover, the crossing relation passes to the limit. Indeed,

$$L_{n_k}(c_{n_k}) = c_{n_k}, \tag{409}$$

and

$$|L^*(c^*) - c^*| \leq |L^*(c^*) - L^*(c_{n_k})| + |L^*(c_{n_k}) - L_{n_k}(c_{n_k})| + |c_{n_k} - c^*|. \tag{410}$$

The first term tends to 0 by continuity of L^*, the second term tends to 0 by uniform convergence, and the third term tends to 0 by construction. Thus

$$L^*(c^*) = c^*. \tag{411}$$

□

By symmetry,

$$L^*(1-c^*) = 1 - L^*(c^*) = 1 - c^*. \tag{412}$$

Thus both c^* and $1-c^*$ are crossing points of L^* with the diagonal. By the one-crossing assumption inherited in the limiting framework, this forces

$$c^* = 1 - c^*, \tag{413}$$

and hence

$$c^* = \frac{1}{2}. \tag{414}$$

Therefore every cluster point of $\{c_n\}_{n\geq 0}$ is equal to $\frac{1}{2}$, and (405) follows.

It is important to note that we can release some of the underlying assumptions.

Remark 83. In the proof of *Claim 62* at the start of the appendix, we relied on the auxiliary assumption

$$c_n \in [\varepsilon, 1-\varepsilon] \qquad (n \geq 0).$$

We now show that, in the regimes in which *Claim 80* applies, this assumption is in fact not needed. The convergence

$$A_n = \int_0^1 |D_n(x)|\, dx \longrightarrow 0, \qquad D_n(x) = L_n(x) + L_n(1-x) - 1. \tag{415}$$

was proved in *Claim 80* without using *Claim 62*, *Arzelà–Ascoli* compactness, or uniform subsequential convergence of L_n. Indeed, its proof uses only the one-sign structure of D_n, the monotonicity properties of D_n, *Helly selection* for the auxiliary monotone functions, and dominated convergence. Therefore (415) may be used before the *uniform Lipschitz* and *Arzelà–Ascoli* compactness arguments.

Claim 84. *Under the hypotheses of Claim 80, the normalizing constants*

$$I_n = \int_0^1 T_n(u)(1 - T_n(u))\, du, \qquad T_n = L_n^{-1}, \tag{416}$$

are uniformly bounded away from zero. That is, there exists a constant $C > 0$ such that

$$I_n \geq C > 0 \qquad \text{for all } n \geq 0. \tag{417}$$

Consequently, in the situation covered by Claim 80, the assumption (239) *in Claim 62 may be omitted.*

Proof. Suppose, to the contrary, that the conclusion is false. Then there exists a subsequence, still denoted by n_k, such that

$$I_{n_k} \to 0. \tag{418}$$

Let μ_n be the probability measure on $[0, 1]$ with distribution function L_n. Since $[0, 1]$ is compact, *Helly's selection principle*, or equivalently *Prokhorov compactness* (see, e.g., [5, Chapter 1] and [34, Chapter 23]) on a compact metric space, gives a further subsequence, again denoted by n_k, and a probability measure μ^* on $[0, 1]$ such that

$$\mu_{n_k} \Rightarrow \mu^*. \tag{419}$$

Let L^* be the distribution function of μ^*. Then

$$L_{n_k}(x) \to L^*(x) \tag{420}$$

at every continuity point x of L^*.

We next identify the possible weak limit. Since $T_n = L_n^{-1}$ is the quantile function corresponding to μ_n, we have

$$I_n = \int_0^1 T_n(u)(1 - T_n(u))\, du = \int_{[0,1]} x(1-x)\, d\mu_n(x). \tag{421}$$

The function $x \mapsto x(1-x)$ is continuous and bounded on $[0, 1]$. Therefore, from (419),

$$\int_{[0,1]} x(1-x)\, d\mu_{n_k}(x) \longrightarrow \int_{[0,1]} x(1-x)\, d\mu^*(x). \tag{422}$$

Together with (418), this gives

$$\int_{[0,1]} x(1-x)\, d\mu^*(x) = 0. \tag{423}$$

Since $x(1-x) > 0$ for $x \in (0, 1)$, the measure μ^* is supported on the two endpoint points $\{0, 1\}$. Hence there exists $p \in [0, 1]$ such that

$$\mu^* = p\delta_0 + (1-p)\delta_1. \tag{424}$$

Equivalently,

$$L^*(x) = \begin{cases} 0, & x < 0, \\ p, & 0 \le x < 1, \\ 1, & x \ge 1. \end{cases} \tag{425}$$

In particular,

$$L^*(x) = p \qquad \text{for every } x \in (0,1). \tag{426}$$

We now use the vanishing of the symmetry defect. For every $x \in (0,1)$, both x and $1-x$ are continuity points of L^*, because L^* is constant on $(0,1)$. Hence

$$L_{n_k}(x) \to p, \qquad L_{n_k}(1-x) \to p,$$

and therefore

$$D_{n_k}(x) = L_{n_k}(x) + L_{n_k}(1-x) - 1 \longrightarrow 2p - 1 \qquad (0 < x < 1). \tag{427}$$

By *Fatou's lemma* and (367),

$$\int_0^1 |2p-1|\, dx \le \liminf_{k\to\infty} \int_0^1 |D_{n_k}(x)|\, dx = 0. \tag{428}$$

Thus

$$2p - 1 = 0, \qquad p = \frac{1}{2}. \tag{429}$$

Consequently the only possible denominator-collapsing limit under (367) is

$$\mu^* = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1, \tag{430}$$

or, equivalently,

$$L^*(x) = \frac{1}{2} \qquad \text{for } 0 < x < 1. \tag{431}$$

It remains to show that this is incompatible with the crossing pattern. Let $c_n \in (0,1)$ be the unique interior fixed point from *Lemma 60*, so that

$$L_n(x) < x \quad (0 < x < c_n), \qquad L_n(x) > x \quad (c_n < x < 1). \tag{432}$$

Since $c_{n_k} \in [0,1]$, by passing to a further subsequence if necessary we may assume

$$c_{n_k} \to c \qquad \text{for some } c \in [0,1]. \tag{433}$$

If $c < 1$, choose

$$x \in (\max\{c, 1/2\}, 1).$$

Then $x > c_{n_k}$ for all sufficiently large k. By the crossing pattern,

$$L_{n_k}(x) > x$$

for all sufficiently large k. Passing to the limit and using (431), we obtain

$$\frac{1}{2} \ge x,$$

which contradicts the choice $x > 1/2$.
If $c = 1$, choose $x \in (0, 1/2)$. Then $x < c_{n_k}$ for all sufficiently large k. By the crossing pattern,

$$L_{n_k}(x) < x$$

for all sufficiently large k. Passing to the limit and using (431), we obtain

$$\frac{1}{2} \le x,$$

which contradicts the choice $x < 1/2$.

Both cases lead to contradictions. Hence the assumption (418) is impossible. Therefore

$$\inf_{n\geq 0} I_n > 0.$$

This proves the claim. $\square$

Corollary 85. *In every regime of Appendix C in which Claim 80 applies, the auxiliary assumption* (239) *in Claim 62 is superfluous. More precisely, under the hypotheses of Claim 80, the conclusion of Claim 62 holds without assuming*

$$c_n \in [\varepsilon, 1-\varepsilon].$$

Proof. *Claim 84* gives a constant $C > 0$ such that

$$I_n \geq C > 0 \qquad (n \geq 0).$$

Since

$$0 \leq T_n(x)(1 - T_n(x)) \leq \frac{1}{4},$$

the iteration formula yields

$$0 \leq L'_{n+1}(x) = \frac{T_n(x)(1-T_n(x))}{I_n} \leq \frac{1/4}{C}.$$

The same direct estimate applies to L_0 from its defining formula. Hence the derivative bound and the *Lipschitz* conclusion of *Claim 62* follow without the $c_n \in [\varepsilon, 1-\varepsilon]$ assumption. $\square$

Variation-diminishing structures and classes of starting laws for which the symmetry-defect contraction applies

This section establishes sufficient conditions for the validity of the constant sign assumptions regarding the initial defect. Consequently, these also constitute sufficient conditions for the symmetry of the subsequential limits of L_n about $\frac{1}{2}$.

We first study the number of sign changes of the symmetry defect in a broader context

$$D_n(x) = L_n(x) + L_n(1-x) - 1, \qquad x \in [0,1], \tag{434}$$

and we relate the sign-change dynamics to classical variation-diminishing theory in the sense of Karlin. Then we formulate explicit sufficient conditions on the starting distribution F ensuring either immediate entrance into the one-sign regime or a strict reduction of sign changes at the first iteration.

Definition 86. Let f be a continuous real-valued function on an interval (a, b). We denote by

$$Z(f; (a,b)) \tag{435}$$

the maximal integer $m \geq 0$ for which there exist points

$$a < x_1 < \cdots < x_{m+1} < b \tag{436}$$

such that

$$f(x_i)f(x_{i+1}) < 0, \qquad i = 1, \ldots, m. \tag{437}$$

Thus $Z(f; (a, b))$ is the number of strict sign changes of f on (a, b). Whenever the interval is $(0, \frac{1}{2})$ we write simply

$$Z(f) = Z\big(f; \big(0, \tfrac{1}{2}\big)\big)\,. \tag{438}$$

Remark 87. The variation-diminishing theorem below for totally nonnegative kernels goes back to the classical theory of total positivity; a standard reference is [35], especially the material in Ch. 5. A direct statistical treatment of variation-diminishing transformations is given in [6]. For modern background on totally positive matrices and kernels, see [51].

We start with an introductory lemma for the *Volterra kernel.*

Lemma 88. *Let*

$$K(x,u) = 1_{\{0\le u\le x\le \frac{1}{2}\}}, \qquad (x,u) \in \left[0, \tfrac{1}{2}\right]^2. \tag{439}$$

Then K is a totally nonnegative kernel on $\left[0,\frac{1}{2}\right]^2$. Equivalently, for every $m \ge 1$ and every strictly increasing tuples

$$0 < x_1 < \cdots < x_m \le \frac{1}{2}, \qquad 0 < u_1 < \cdots < u_m \le \frac{1}{2}, \tag{440}$$

the determinant

$$\det(K(x_i,u_j))_{i,j=1}^m \tag{441}$$

is nonnegative. In fact, each such determinant is either 0 *or* 1.

Proof. Fix m and strictly increasing tuples $x_1 < \cdots < x_m$ and $u_1 < \cdots < u_m$. Set

$$A_{ij} = K(x_i,u_j) = 1_{\{u_j \le x_i\}}. \tag{442}$$

For each row index i, the sequence $A_{i1},\dots,A_{im}$ consists of an initial block of 1's followed by a block of 0's, because the u_j are increasing. Hence each row has the form

$$(\underbrace{1,\dots,1}_{k_i \text{ entries}},0,\dots,0) \tag{443}$$

for some integer $k_i \in \{0,\dots,m\}$. Since the x_i are increasing, the row parameters satisfy

$$0 \le k_1 \le k_2 \le \cdots \le k_m \le m. \tag{444}$$

If $k_i = k_{i+1}$ for some i, then rows i and $i+1$ are identical, so

$$\det A = 0. \tag{445}$$

Therefore a nonzero determinant is possible only when

$$k_i = i \qquad \text{for every } i = 1,\dots,m. \tag{446}$$

In that case A is lower triangular with all diagonal entries equal to 1, hence

$$\det A = 1. \tag{447}$$

Thus every minor is either 0 or 1, in particular nonnegative.

Now we can formulate the *Karlin's variation-diminishing theorem* for the *Volterra kernel.* □

Theorem 89. *Let*

$$(\mathcal{K}f)(x) = \int\limits_0^{1/2} K(x,u)f(u)\,du = \int\limits_0^{x} f(u)\,du, \qquad 0 \le x \le \frac{1}{2}, \tag{448}$$

where K is the kernel from Lemma 88. Then

$$Z(\mathcal{K}f) \le Z(f) \tag{449}$$

for every continuous function f on $[0,\frac{1}{2}]$.

Proof. By *Lemma 88*, the kernel K is totally nonnegative. The variation-diminishing theorem for totally nonnegative kernels therefore applies; see *Remark 87*. This yields (449). □

Claim 90. *For every $n \ge 0$, define*

$$S_n(u) = T_n(u) + T_n(1-u) - 1, \qquad 0 \le u \le 1, \tag{450}$$

and

$$G_n(u) = w(T_n(u)) - w(T_n(1-u)), \qquad w(y) = y(1-y), \qquad 0 \le u \le \frac{1}{2}. \tag{451}$$

Then the following hold.

1. *For* $0 \le x \le \frac{1}{2}$,

$$D_{n+1}(x) = \frac{1}{I_n} \int_0^x G_n(u)\, du, \qquad I_n = \int_0^1 w(T_n(v))\, dv > 0. \tag{452}$$

2. *For* $0 < u < \frac{1}{2}$,

$$G_n(u) = (T_n(1-u) - T_n(u)) S_n(u), \tag{453}$$

hence

$$Z(G_n) = Z(S_n). \tag{454}$$

3. *We have*

$$Z(S_n) = Z(D_n). \tag{455}$$

4. *Consequently,*

$$Z(D_{n+1}) \le Z(D_n) \qquad \textit{for every } n \ge 0. \tag{456}$$

Proof. For the first statement, recall that

$$L_{n+1}(x) = \frac{\int_0^x w(T_n(u))\, du}{I_n}. \tag{457}$$

Hence for $0 \le x \le \frac{1}{2}$,

$$\begin{aligned} D_{n+1}(x) &= L_{n+1}(x) + L_{n+1}(1-x) - 1 \\ &= \frac{1}{I_n} \left(\int_0^x w(T_n(u))\, du + \int_0^{1-x} w(T_n(u))\, du - \int_0^1 w(T_n(u))\, du \right) \\ &= \frac{1}{I_n} \left(\int_0^x w(T_n(u))\, du - \int_{1-x}^1 w(T_n(u))\, du \right). \end{aligned} \tag{458}$$

Changing variables $v = 1 - u$ in the second integral, we obtain

$$D_{n+1}(x) = \frac{1}{I_n} \int_0^x (w(T_n(u)) - w(T_n(1-u)))\, du, \tag{459}$$

which is (452).

For the second statement, use

$$w(a) - w(b) = (a-b)(1-a-b). \tag{460}$$

With $a = T_n(u)$ and $b = T_n(1-u)$ this gives

$$\begin{aligned} G_n(u) &= w(T_n(u)) - w(T_n(1-u)) \\ &= (T_n(u) - T_n(1-u))(1 - T_n(u) - T_n(1-u)) \\ &= (T_n(1-u) - T_n(u))(T_n(u) + T_n(1-u) - 1), \end{aligned} \tag{461}$$

which is exactly (453). Since T_n is strictly increasing, the factor $T_n(1-u) - T_n(u)$ is strictly positive on $(0, \frac{1}{2})$, so G_n and S_n have the same sign pattern there. Hence (454) holds.

For the third statement, let $x \in (0,1)$ and set $u = L_n(x)$. Then

$$S_n(u) = T_n(u) + T_n(1-u) - 1 = x + T_n(1 - L_n(x)) - 1. \tag{462}$$

Therefore

$$S_n(L_n(x)) = T_n(1 - L_n(x)) - (1 - x). \tag{463}$$

Because T_n is increasing, the sign of this quantity is the sign of

$$(1 - L_n(x)) - L_n(1 - x) = -(L_n(x) + L_n(1 - x) - 1) = -D_n(x). \tag{464}$$

Thus

$$\operatorname{sgn}(S_n(L_n(x))) = -\operatorname{sgn}(D_n(x)) \qquad (x \in (0,1),\ D_n(x) \neq 0). \tag{465}$$

Since L_n is a strictly increasing homeomorphism of $[0,1]$, composition with L_n preserves the number of sign changes. Hence (455) follows.

Finally, by *Theorem 89* and (452),

$$Z(D_{n+1}) \leq Z(G_n). \tag{466}$$

Using (454) and (455) we conclude that

$$Z(D_{n+1}) \leq Z(D_n), \tag{467}$$

which is (456). □

Remark 91. The preceding argument gives only the non-strict inequality

$$Z(D_{n+1}) \leq Z(D_n). \tag{468}$$

It does not give the strict inequality

$$Z(D_{n+1}) < Z(D_n) \qquad \text{whenever } Z(D_n) > 0. \tag{469}$$

There are two reasons for this.

First, the *Volterra kernel* $K(x,u) = 1_{\{u \leq x\}}$ is totally nonnegative, but it is not strictly totally positive. Indeed, for instance, if

$$0 < u_1 < u_2 < x_1 < x_2 < \frac{1}{2}, \tag{470}$$

then

$$\det \begin{pmatrix} K(x_1,u_1) & K(x_1,u_2) \\ K(x_2,u_1) & K(x_2,u_2) \end{pmatrix} = \det \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = 0. \tag{471}$$

Thus the strict versions of *Karlin's theorems* do not apply directly to this kernel.

Second, strict loss of sign changes is false in general for the *Volterra operator* even on smooth inputs. For example, on $[0, \frac{1}{2}]$ define

$$f(u) = u - \frac{1}{8}. \tag{472}$$

Then f is C^∞ and has exactly one sign change:

$$Z(f) = 1. \tag{473}$$

However

$$(\mathcal{K}f)(x) = \int_0^x \left(u - \frac{1}{8}\right) du = \frac{x^2}{2} - \frac{x}{8} = x\left(\frac{x}{2} - \frac{1}{8}\right), \tag{474}$$

which also has exactly one sign change on $(0, \frac{1}{2})$. Hence

$$Z(\mathcal{K}f) = 1 = Z(f). \tag{475}$$

Therefore strict reduction of sign changes cannot be deduced from the kernel alone. The weight $w(y) = y(1-y)$ does not alter this conclusion at the level of the *Karlin argument*, because w enters only through the input profile

$$G_n(u) = w(T_n(u)) - w(T_n(1-u)), \tag{476}$$

while the transform from G_n to D_{n+1} remains the same *Volterra operator*.

Claim 92. *Let f be a continuous function on $[0, a]$, $a > 0$, with finitely many sign changes on $(0, a)$. Suppose that the zeros of f on $[0, a]$ satisfy*

$$0 = \xi_0 < \xi_1 < \cdots < \xi_m < \xi_{m+1} = a,$$

and that f has constant sign on each nodal interval

$$I_k = (\xi_{k-1}, \xi_k), \qquad k = 1, \ldots, m+1,$$

with alternating signs from one interval to the next.

Define the nodal masses

$$M_k = \int_{\xi_{k-1}}^{\xi_k} f(u)\,du, \qquad k = 1, \ldots, m+1. \tag{477}$$

Assume moreover that

$$|M_1| \geq |M_2| \geq \cdots \geq |M_{m+1}|. \tag{478}$$

Then the primitive

$$F(x) = \int_0^x f(u)\,du$$

satisfies

$$Z(F) < Z(f). \tag{479}$$

Proof. Because the signs of f alternate from lobe to lobe, the numbers M_k alternate in sign as well. Suppose, for definiteness, that $M_1 > 0$; the other case is identical after multiplying by -1. Then

$$M_1 > 0, \quad M_2 < 0, \quad M_3 > 0, \ \ldots$$

and the monotonicity (478) implies that the partial sums

$$S_k = M_1 + \cdots + M_k, \qquad k = 1, \ldots, m+1,$$

are all nonnegative. More precisely, each S_k has the sign of M_1 unless an equality of consecutive absolute nodal masses produces a zero partial sum. In either case, the sequence $\{S_k\}_{k\geq 0}$ cannot alternate signs.

Now $F(\xi_k) = S_k$. Since the values $F(\xi_k)$ do not alternate signs, the primitive F cannot alternate sign across all nodal intervals of f. Therefore F must have strictly fewer sign changes than f, proving (479). □

Corollary 93. *Fix $n \geq 0$. Let the zeros of*

$$G_n(u) = w(T_n(u)) - w(T_n(1-u)) \tag{480}$$

on $(0, \frac{1}{2})$ be

$$0 = \xi_0 < \xi_1 < \cdots < \xi_m < \xi_{m+1} = \frac{1}{2}, \tag{481}$$

and define the corresponding nodal masses

$$M_k^{(n)} = \int_{\xi_{k-1}}^{\xi_k} G_n(u)\,du, \qquad k = 1, \ldots, m+1. \tag{482}$$

If

$$|M_1^{(n)}| \geq |M_2^{(n)}| \geq \cdots \geq |M_{m+1}^{(n)}|, \tag{483}$$

then

$$Z(D_{n+1}) < Z(D_n). \tag{484}$$

Proof. By *Claim 90*,

$$D_{n+1}(x) = \frac{1}{I_n} \int_0^x G_n(u)\,du \qquad \text{on } \left[0, \frac{1}{2}\right], \tag{485}$$

and

$$Z(G_n) = Z(D_n). \tag{486}$$

Since multiplication by the positive constant I_n^{-1} does not affect sign changes, *Claim 92* applied to $f = G_n$ yields

$$Z(D_{n+1}) < Z(G_n) = Z(D_n). \tag{487}$$

□

The preceding corollary is not, by itself, a condition on the starting law. Rather, it is a stepwise criterion: whenever the nodal masses of the current profile G_n are ordered as in (478), the next iterate loses at least one sign change. Thus, if this criterion can be verified at every iterate before the one-sign regime is reached, then the process enters the one-sign regime after finitely many steps. If the condition fails at some iterate, the corollary gives no conclusion at that step.

Claim 94. *Let $Q = F^{-1}$ be the quantile function of the starting distribution F on $[0,1]$, and define on $[0, \frac{1}{2}]$*

$$H_F(u) = Q(u) + Q(1-u) - 1, \tag{488}$$

$$W_F(u) = Q(1-u) - Q(u), \tag{489}$$

and

$$I_F = \int_0^1 Q(t)(1 - Q(t))\,dt. \tag{490}$$

Assume that $I_F > 0$. Then
1. $W_F(u) \geq 0$ *and W_F is nonincreasing on $[0, \frac{1}{2}]$;*
2. *the initial defect*

$$D_0(x) = L_0(x) + L_0(1-x) - 1 \tag{491}$$

satisfies

$$D_0(x) = \frac{1}{I_F} \int_0^x W_F(u)\,H_F(u)\,du, \qquad 0 \leq x \leq \frac{1}{2}; \tag{492}$$

3. *if, in addition, Q is continuous and strictly increasing on $[0,1]$, then*

$$Z(D_0) \leq Z(H_F). \tag{493}$$

Proof. Since Q is increasing,

$$W_F(u) = Q(1-u) - Q(u) \geq 0 \qquad \left(0 \leq u \leq \frac{1}{2}\right). \tag{494}$$

If $0 \leq u < v \leq \frac{1}{2}$, then

$$Q(1-u) \geq Q(1-v), \qquad Q(u) \leq Q(v), \tag{495}$$

so

$$W_F(u) = Q(1-u) - Q(u) \geq Q(1-v) - Q(v) = W_F(v). \tag{496}$$

Thus W_F is nonincreasing.

Next, by the definition of L_0,

$$L_0(x) = \frac{\int_0^x Q(t)(1 - Q(t))\,dt}{I_F}. \tag{497}$$

Exactly as in the proof of *Claim 90*, for $0 \le x \le \frac{1}{2}$,

$$D_0(x) = \frac{1}{I_F}\left(\int_0^x Q(u)(1-Q(u))\,du - \int_{1-x}^1 Q(u)(1-Q(u))\,du\right)$$
$$= \frac{1}{I_F}\int_0^x (Q(u)(1-Q(u)) - Q(1-u)(1-Q(1-u)))\,du. \tag{498}$$

Using

$$a(1-a) - b(1-b) = (b-a)(a+b-1), \tag{499}$$

with $a = Q(u)$ and $b = Q(1-u)$, we obtain

$$Q(u)(1-Q(u)) - Q(1-u)(1-Q(1-u)) = (Q(1-u) - Q(u))(Q(u) + Q(1-u) - 1), \tag{500}$$

which proves (492).

Assume now that Q is continuous and strictly increasing on $[0,1]$. Then H_F and W_F are continuous on $[0, \frac{1}{2}]$, and

$$W_F(u) > 0 \qquad \text{for every } 0 < u < \frac{1}{2}. \tag{501}$$

Hence $W_F H_F$ is continuous on $[0, \frac{1}{2}]$, and by *Theorem 89* applied to

$$D_0(x) = \frac{1}{I_F}\int_0^x (W_F H_F)(u)\,du, \tag{502}$$

we obtain

$$Z(D_0) \le Z(W_F H_F). \tag{503}$$

Since $W_F(u) > 0$ on $(0, \frac{1}{2})$, the functions $W_F H_F$ and H_F have the same sign changes on $(0, \frac{1}{2})$, and therefore

$$Z(W_F H_F) = Z(H_F). \tag{504}$$

This proves (493). □

Corollary 95. *If*

$$H_F(u) = Q(u) + Q(1-u) - 1 \tag{505}$$

has a constant sign on $[0, \frac{1}{2}]$, then D_0 has a constant sign on $[0,1]$. Consequently, by Lemma 76, all later defects D_n have a constant sign on $[0,1]$, with alternating sign in n.

Proof. If H_F has a constant sign, then so does $W_F H_F$ because $W_F \ge 0$ on $[0, \frac{1}{2}]$. Formula (492) then shows that D_0' has a fixed sign on $[0, \frac{1}{2}]$. Since $D_0(0) = 0$ and $D_0(1-x) = D_0(x)$, it follows that D_0 has a constant sign on $[0,1]$. The persistence statement is then exactly *Lemma 76*. □

Corollary 96. *Assume that Q is continuous and strictly increasing on $[0,1]$. Assume that H_F has exactly one zero*

$$0 < \xi < \frac{1}{2} \tag{506}$$

on $(0, \frac{1}{2})$. If

$$\left|\int_0^\xi W_F(u)H_F(u)\,du\right| \ge \left|\int_\xi^{1/2} W_F(u)H_F(u)\,du\right|, \tag{507}$$

then D_0 has a constant sign on $[0,1]$.

A simpler sufficient condition is

$$\int_0^\xi |H_F(u)|\,du \ge \int_\xi^{1/2} |H_F(u)|\,du, \tag{508}$$

because W_F is nonincreasing on $[0, \frac{1}{2}]$.

Proof. If H_F has exactly one zero, then $W_F H_F$ has exactly two nodal intervals on $(0, \frac{1}{2})$ because $W_F(u) > 0$ for every $0 < u < \frac{1}{2}$. Condition (507) says exactly that the first nodal mass dominates the second in absolute value. Therefore *Claim 92* implies that the primitive (492) has no sign change on $(0, \frac{1}{2})$, i.e. D_0 has a constant sign there, and hence on all of $[0, 1]$ by symmetry.

To obtain (508) as a sufficient condition, note that W_F is nonincreasing on $[0, \frac{1}{2}]$. Thus

$$\left| \int_0^{\xi} W_F(u) H_F(u)\, du \right| \geq W_F(\xi) \int_0^{\xi} |H_F(u)|\, du, \tag{509}$$

whereas

$$\left| \int_{\xi}^{1/2} W_F(u) H_F(u)\, du \right| \leq W_F(\xi) \int_{\xi}^{1/2} |H_F(u)|\, du. \tag{510}$$

Hence (508) implies (507). □

Corollary 97. *Assume that Q is continuous and strictly increasing on $[0, 1]$. Assume that the zeros of H_F on $(0, \frac{1}{2})$ are*

$$0 = \xi_0 < \xi_1 < \cdots < \xi_m < \xi_{m+1} = \frac{1}{2}, \tag{511}$$

and define the lobe areas

$$A_k(F) = \int_{\xi_{k-1}}^{\xi_k} |H_F(u)|\, du, \qquad k = 1, \ldots, m + 1. \tag{512}$$

If

$$A_1(F) \geq A_2(F) \geq \cdots \geq A_{m+1}(F), \tag{513}$$

then

$$Z(D_0) < Z(H_F). \tag{514}$$

Proof. Let

$$M_k(F) = \int_{\xi_{k-1}}^{\xi_k} W_F(u) H_F(u)\, du \tag{515}$$

be the weighted nodal masses of $W_F H_F$. Since W_F is nonincreasing on $[0, \frac{1}{2}]$, we have

$$|M_k(F)| = \int_{\xi_{k-1}}^{\xi_k} W_F(u) |H_F(u)|\, du \geq W_F(\xi_k) \int_{\xi_{k-1}}^{\xi_k} |H_F(u)|\, du = W_F(\xi_k) A_k(F), \tag{516}$$

and also

$$|M_{k+1}(F)| \leq W_F(\xi_k) \int_{\xi_k}^{\xi_{k+1}} |H_F(u)|\, du = W_F(\xi_k) A_{k+1}(F). \tag{517}$$

Therefore (512) implies

$$|M_1(F)| \geq |M_2(F)| \geq \cdots \geq |M_{m+1}(F)|. \tag{518}$$

Applying *Claim 92* to

$$f(u) = W_F(u) H_F(u), \tag{519}$$

and using (492), yields

$$Z(D_0) < Z(W_F H_F) = Z(H_F), \tag{520}$$

because $W_F(u) > 0$ for every $0 < u < \frac{1}{2}$. □

Claim 98. *Assume that F is continuous and strictly increasing on $[0,1]$, so that $Q = F^{-1}$ is its genuine inverse. Let*

$$F^{\sharp}(x) = 1 - F(1-x), \qquad x \in [0,1], \tag{521}$$

be the cdf of the reflected random variable $1 - X$ when $X \sim F$. Then the following are equivalent:

1.
$$Q(u) + Q(1-u) - 1 \le 0 \qquad \forall u \in [0,1]; \tag{522}$$

2.
$$F(x) \ge F^{\sharp}(x) = 1 - F(1-x) \qquad \forall x \in [0,1]; \tag{523}$$

3.
$$X \preceq_{\mathrm{st}} 1 - X, \tag{524}$$

that is, X is stochastically dominated by its reflection.

Likewise,

$$Q(u) + Q(1-u) - 1 \ge 0 \ \forall u \tag{525}$$

is equivalent to

$$F(x) \le 1 - F(1-x) \ \forall x, \tag{526}$$

that is, $1 - X \preceq_{\mathrm{st}} X$.

Proof. We prove the first equivalence; the second is identical with all inequalities reversed.

Assume first that

$$Q(u) + Q(1-u) - 1 \le 0 \qquad \forall u \in [0,1]. \tag{527}$$

Fix $x \in [0,1]$ and put $u = F(x)$. Then $x = Q(u)$, and the assumed inequality gives

$$Q(1-u) \le 1 - Q(u) = 1 - x. \tag{528}$$

Applying the increasing map F to both sides, we obtain

$$1 - u \le F(1-x). \tag{529}$$

Since $u = F(x)$, this is

$$F(x) \ge 1 - F(1-x) = F^{\sharp}(x). \tag{530}$$

Conversely, assume

$$F(x) \ge 1 - F(1-x) \qquad \forall x \in [0,1]. \tag{531}$$

Fix $u \in [0,1]$ and put $x = Q(u)$. Then $u = F(x)$, so

$$u \ge 1 - F(1-x). \tag{532}$$

Hence

$$1 - u \le F(1-x). \tag{533}$$

Applying the increasing map $Q = F^{-1}$,

$$Q(1-u) \le 1 - x = 1 - Q(u), \tag{534}$$

that is,

$$Q(u) + Q(1-u) - 1 \le 0. \tag{535}$$

This proves the equivalence.

The equivalence with stochastic order is the standard definition:

$$X \preceq_{\mathrm{st}} Y \quad \Longleftrightarrow \quad F_X(x) \ge F_Y(x) \ \forall x. \tag{536}$$

With $Y = 1 - X$, the cdf of Y is exactly $F^{\sharp}$. $\square$

Corollary 99. *Assume that F has a continuous positive density f on $(0,1)$, and define*

$$G_F(x) = F(x) + F(1-x) - 1, \qquad x \in [0,1]. \tag{537}$$

Then:

1. *If $f(x) - f(1-x)$ changes sign at most once on $(0,1)$, then G_F has a constant sign on $[0,1]$.*
2. *Consequently, by Claim 98,*
$$H_F(u) = Q(u) + Q(1-u) - 1 \tag{538}$$
has a constant sign on $[0,1]$, and by Corollary 95 the first defect D_0 has a constant sign on $[0,1]$.
3. *A sufficient condition for the hypothesis in (1) is that the ratio*
$$x \longmapsto \frac{f(x)}{f(1-x)} \tag{539}$$
be monotone on $(0,1)$.

Proof. Since

$$G_F(0) = 0, \qquad G_F(1) = 0, \tag{540}$$

and

$$G'_F(x) = f(x) - f(1-x), \tag{541}$$

the first claim is immediate: if G'_F changes sign at most once, then G_F is first monotone and then monotone in the opposite direction; since it vanishes at both endpoints, it cannot cross the horizontal axis in the interior, and therefore has a constant sign on $[0,1]$.

The second statement then follows from *Claim 98* and *Corollary 95.*

For the third statement, observe that

$$f(x) - f(1-x) = f(1-x)\left(\frac{f(x)}{f(1-x)} - 1\right). \tag{542}$$

Since $f(1-x) > 0$, the sign changes of $f(x) - f(1-x)$ are exactly the sign changes of $\frac{f(x)}{f(1-x)} - 1$. If the ratio is monotone, then this difference changes sign at most once. □

Claim 100. *Let $Q = F^{-1}$ and H_F, W_F be as in Claim 94. The following conditions give explicit sufficient hypotheses on the starting law F: items (i), (ii) and (iv) imply that the initial defect D_0 has a constant sign on $[0,1]$, whereas item (iii) gives the strict initial sign-change reduction*

$$Z(D_0) < Z(H_F). \tag{543}$$

1. ***Immediate one-sign regime.*** *If*
$$H_F(u) = Q(u) + Q(1-u) - 1 \tag{544}$$
has a constant sign on $[0, \frac{1}{2}]$, then D_0 has a constant sign on $[0,1]$.
2. ***Two-lobe dominance.*** *Assume moreover that Q is continuous and strictly increasing on $[0,1]$. Suppose H_F has exactly one zero*
$$0 < \xi < \frac{1}{2} \tag{545}$$
on $(0, \frac{1}{2})$. If
$$\left|\int_0^{\xi} W_F(u) H_F(u)\, du\right| \geq \left|\int_{\xi}^{1/2} W_F(u) H_F(u)\, du\right|, \tag{546}$$
then D_0 has a constant sign on $[0,1]$. A simpler sufficient condition is
$$\int_0^{\xi} |H_F(u)|\, du \geq \int_{\xi}^{1/2} |H_F(u)|\, du. \tag{547}$$

3. ***General decreasing-lobe criterion.*** *Assume moreover that Q is continuous and strictly increasing on $[0,1]$. Suppose the zeros of H_F on $(0,\frac{1}{2})$ are*

$$0=\xi_0<\xi_1<\cdots<\xi_m<\xi_{m+1}=\frac{1}{2}, \tag{548}$$

and define the lobe areas

$$A_k(F)=\int_{\xi_{k-1}}^{\xi_k}|H_F(u)|\,du, \qquad k=1,\dots,m+1. \tag{549}$$

If

$$A_1(F)\ge A_2(F)\ge\cdots\ge A_{m+1}(F), \tag{550}$$

then

$$Z(D_0)<Z(H_F). \tag{551}$$

4. ***Density-level criterion.*** *If F has a continuous positive density f on $(0,1)$ and the function*

$$x\mapsto f(x)-f(1-x) \tag{552}$$

changes sign at most once on $(0,1)$, then H_F has a constant sign on $[0,1]$, hence D_0 has a constant sign on $[0,1]$. A sufficient condition for this is that

$$x\longmapsto \frac{f(x)}{f(1-x)} \tag{553}$$

be monotone on $(0,1)$.

Proof. Statement (i) is exactly *Corollary 95*.

Statement (ii) is exactly *Corollary 96*.

Statement (iii) is exactly *Corollary 97*.

For (iv), define

$$G_F(x)=F(x)+F(1-x)-1, \qquad x\in[0,1]. \tag{554}$$

Then

$$G_F(0)=0, \qquad G_F(1)=0, \qquad G_F'(x)=f(x)-f(1-x). \tag{555}$$

If G_F' changes sign at most once, then G_F is first monotone and then monotone in the opposite direction; since it vanishes at both endpoints, it cannot cross the horizontal axis in the interior, and therefore has a constant sign on $[0,1]$.

By Claim *98*, this is equivalent to $H_F(u)=Q(u)+Q(1-u)-1$ having a constant sign on $[0,1]$, and thus on $[0,\frac{1}{2}]$. Hence (i) applies and yields that D_0 has a constant sign.

Finally, if

$$x\longmapsto \frac{f(x)}{f(1-x)} \tag{556}$$

is monotone, then

$$f(x)-f(1-x)=f(1-x)\left(\frac{f(x)}{f(1-x)}-1\right) \tag{557}$$

changes sign at most once, because $f(1-x)>0$. Thus the first part of (iv) applies. □

Remark 101. The sufficient conditions of *Claim 100* admit the following interpretation.

1. The strongest condition is the one-sign condition on

$$H_F(u)=F^{-1}(u)+F^{-1}(1-u)-1. \tag{558}$$

Under the regularity assumptions of *Claim 98*, this is equivalent to a global stochastic ordering between the starting law and its reflection.

2. The two-lobe condition is the sharpest practically useful criterion when H_F changes sign exactly once on $(0, \frac{1}{2})$. It says that the first lobe, after weighting by
$$W_F(u) = F^{-1}(1-u) - F^{-1}(u), \tag{559}$$
dominates the second one. Because W_F is decreasing, the simpler unweighted area comparison
$$\int_0^{\xi} |H_F(u)|\, du \geq \int_{\xi}^{1/2} |H_F(u)|\, du \tag{560}$$
already suffices.
3. In the multiple-lobe case, the monotone decrease of the unweighted lobe areas of H_F implies a monotone decrease of the weighted nodal masses of $W_F H_F$, and this is enough to force strict loss of sign changes at the first step.
4. At the density level, one-sided skewness relative to the reflection, as expressed by the one-crossing property of
$$f(x) - f(1-x), \tag{561}$$
or more concretely by monotonicity of the ratio
$$f(x)/f(1-x), \tag{562}$$
is sufficient to enter the one-sign regime immediately.

These conditions are all sufficient, not necessary. They are intended as checkable hypotheses on the starting law F which guarantee that the process either begins in the one-sign regime or strictly reduces the number of sign changes at the first step.

Remark 102. The results above establish the following rigorous chain.
1. The defect iteration is variation diminishing in the sense that
$$Z(D_{n+1}) \leq Z(D_n) \qquad \forall n \geq 0. \tag{563}$$
2. Strict reduction of sign changes does not follow from Karlin theory alone, because the *Volterra kernel* is totally nonnegative but not strictly totally positive.
3. Nevertheless, strict reduction can be forced by explicit nodal-mass conditions, either at a general iterate via *Corollary 93*, or directly at the first step from the starting law via *Claim 100*.
4. If the process reaches the one-sign regime at some step N, then the same argument as in *Lemma 76* implies that all subsequent defects D_n, $n \geq N$, have a constant sign, alternating in n, and therefore the scalar symmetry defect
$$A_n = \int_0^1 |D_n(x)|\, dx \tag{564}$$
is available for the contraction analysis developed above.

What remains open, in full generality, is a theorem ensuring that every starting law enters the one-sign regime after finitely many steps. The results of the present subsection provide rigorous sufficient conditions on F for that entrance to occur immediately or after a provable strict reduction of sign changes at the first step.

Properties of the compound maps $\Phi_n(x)$

We consider now the compound maps built from the inverses:
$$\Phi_n(x) = L_0^{-1} \circ L_1^{-1} \circ \cdots \circ L_n^{-1}(x), \qquad x \in [0,1] \tag{565}$$

and list several important properties of them.

Each Φ_n is a continuous strictly increasing self-map of $[0,1]$. The following single-crossing structure for Φ_n is inherited from that of the L_n.

Lemma 103. *For each $n \geq 0$, the function $\Phi_n(x)$has two fixed points at 0 and 1, and a unique fixed point in $(0,1)$. If we denote the latter by ϕ_n then*

$$\Phi_n(x) > x \quad \text{for } x \in (0, \phi_n), \qquad \Phi_n(x) < x \quad \text{for } x \in (\phi_n, 1). \tag{566}$$

Moreover, there exists a closed interval $I_n \subset (0,1)$ such that $\phi_n \in I_n$ and

$$\Phi_n^{'}(x) \leq 1 \quad \text{for all } x \in I_n. \tag{567}$$

Proof. We can notice that similar behavior to the one described in this claim holds for the inverse $L_n^{-1}(x)$ in terms of the crossing and its general shape by considering the previous claim and the fact that $L_n^{-1}(x)$ is a mirror image of $L_n(x)$ with respect to the diagonal x. The current claim essentially says that the composite function $\Phi_n(x)$ inherits very generally behavior of all the inverses $L_n(x)$. It is not at all obvious how exactly this should happen, and the proof below serves for that purpose.

We will make the proof by induction. Initially, we consider the composition $L^{0,-1}(L^{1,-1}(x))$. We introduce now the deviations $\overline{h}_i(x) = x - L^{i,-1}(x)$ for $i = 0,1$ and the one for the composite function $\overline{h}_{\Phi_1}(x) = x - \Phi_1(x) = x - L^{0,-1}(L^{1,-1}(x))$. We will also keep to the notation for the respective fixed points as well as their properties from the previous claims.

First, we will prove that $\Phi_1(x)$ has a unique fixed point in $(0,1)$. We notice that we can write

$$\overline{h}_{\Phi_1}(x) = x - L^{0,-1}(L^{1,-1}(x)) = x - L^{1,-1}(x) + L^{1,-1}(x) - L^{0,-1}(L^{1,-1}(x)) = \overline{h}_1(x) + \overline{h}_0(L^{1,-1}(x)). \tag{568}$$

We can define now $\overline{c}_0$ by $\overline{c}_0 = L_1(c_0)$. Then for $\overline{h}_1$ we have:

$$\overline{h}_0(L^{1,-1}(x)) > 0 \quad \text{for } L^{1,-1}(x) > c_0 \quad \iff \quad \overline{h}_0(L^{1,-1}(x)) > 0 \quad \text{for } x \in (\overline{c}_0, 1), \tag{569}$$

and

$$\overline{h}_0(L^{1,-1}(x)) < 0 \quad \text{for } L^{1,-1}(x) < c_0 \quad \iff \quad \overline{h}_0(L^{1,-1}(x)) < 0 \quad \text{for } x \in (0, \overline{c}_0). \tag{570}$$

Without loss of generality, we can assume that the unique zero of $\overline{h}_1$ satisfies $c_1 < \overline{c}_0$, i.e., $c_1 < L_1(c_0)$ (the proof for the other case is analogous).

Now we have the following analysis for $\overline{h}_{\Phi_1}(x) = \overline{h}_1(x) + \overline{h}_0(L^{1,-1}(x))$ depending on x :

- **On** $(0, c_1)$**:** Since $\overline{h}_1(x)$ is negative for $x < c_1$ by the crossing property, and since $\overline{h}_0(L^{1,-1}(x))$ is also negative by the crossing property (because $x < c_1 < \overline{c}_0$ implies $L^{1,-1}(x) < L^{1,-1}(c_1) \leq L^{1,-1}(\overline{c}_0) = c_0$), the sum of these two terms must be negative. Therefore,$\overline{h}_{\Phi_1}(x) = \overline{h}_1(x) + \overline{h}_0(L^{1,-1}(x)) < 0$;
- **At** $x = c_1$**:** By definition, $\overline{h}_1(c_1) = 0$. Since $c_1 < \overline{c}_0$ implies $L^{1,-1}(c_1) < c_0$, we have $\overline{h}_0(L^{1,-1}(x)) < 0$. Thus, $\overline{h}_{\Phi_1}(x) = \overline{h}_1(x) + \overline{h}_0(L^{1,-1}(x)) < 0$;
- **On** $(c_1, \overline{c}_0)$**:** In this interval, the two components of $\overline{h}_{\Phi_1}(x)$ have opposing signs. Since $x > c_1$, we have $\overline{h}_1(x) > 0$; conversely, since $x < \overline{c}_0$, it follows that $\overline{h}_0(L^{1,-1}(x)) < 0$. The existence of a root is guaranteed by the *Intermediate Value Theorem*. As established previously, $h_{\Phi_1}(c_1) < 0$. Furthermore, $h_{\Phi_1}(\overline{c}_0) > 0$ (as we will show in the next section). Given this change in sign, there must be at least one point $\phi_1 \in (c_1, \overline{c}_0)$ such that $\overline{h}_{\Phi_1}(\phi_1) = 0$. The uniqueness of this root follows from the strict monotonicity of $\overline{h}_{\Phi_1}(x)$, a property inherited from $\overline{h}_0$ and $\overline{h}_1$. At this unique point ϕ_1, the positive contribution from $\overline{h}_1$ is precisely balanced by the negative contribution from $\overline{h}_0$;
- **At** $x = \overline{c}_0$**:** By definition, $\overline{h}_0(\overline{c}_0) = 0$. Also, for $x > c_1$ we already have $\overline{h}_1(x) > 0$. Thus, $\overline{h}_{\Phi_1}(x) = \overline{h}_1(x) + \overline{h}_0(L^{1,-1}(x)) > 0$;
- **On** $(\overline{c}_0, 1)$**:** For $x > \overline{c}_0$, we have $L^{1,-1}(x) > c_0$ and hence $\overline{h}_0(L^{1,-1}(x)) > 0$. Also, for $x > c_1$ we already have $\overline{h}_1(x) > 0$. Therefore, $\overline{h}_{\Phi_1}(x) = \overline{h}_1(x) + \overline{h}_0(L^{1,-1}(x)) > 0$.

Second, we will prove that $\Phi_1(x) > x$ holds in the interval $(0, \phi_1)$ and $\Phi_1(x) < x$ holds in the interval $(\phi_1, 1)$. For that purpose, we want to show that the equation $\overline{h}_{\Phi_1}^{'}(x) = 0$ has exactly two solutions in $(0,1)$.

Regarding the existence, we already know from above that $\overline{h}_{\Phi_1}(\phi_1) = 0$. This allows to have:

- On the interval $[0,\phi_1]$, since $\overline{h}_{\Phi_1}(0)=0$ and $\overline{h}_{\Phi_1}(\phi_1)=0$, the *Rolle's Theorem* guarantees the existence of some $\phi_1^{(1)}\in(0,\phi_1)$ such that $\overline{h}'_{\Phi_1}(\phi_1^{(1)})=0$;
- Similarly, on the interval $[\phi_1,1]$, since $\overline{h}_{\Phi_1}(\phi_1)=0$ and $\overline{h}_{\Phi_1}(1)=0$, there exists $\phi_1^{(2)}\in(\phi_1,1)$ such that $\overline{h}'_{\Phi_1}(\phi_1^{(2)})=0$.

Regarding the uniqueness, we use a contradiction argument. Take first the interval $(0,\phi_1)$. Let's assume that there exist two distinct points a and b in it such that $\overline{h}'_{\Phi_1}(a)=\overline{h}'_{\Phi_1}(b)=0$ holds. Then again by the *Rolle's Theorem*, there is a $c\in(a,b)$ with $\overline{h}''_{\Phi_1}(c)=0$. But we know that $\overline{h}'_{\Phi_1}(x)=\overline{h}'_1(x)+\overline{h}'_0(L^{1,-1}(x))\,[L^{1,-1}(x)]'$ and also $\overline{h}''_{\Phi_1}(x)=\overline{h}''_1(x)+\overline{h}''_0(L^{1,-1}(x))\,\left[[L^{1,-1}(x)]'\right]^2+\overline{h}'_0(L^{1,-1}(x))\,[L^{1,-1}(x)]''$. Since each deviation function $\overline{h}_0$ and $\overline{h}_1$ has exactly one local maximum on $(0,c_0)$ and $(0,c_1)$, their derivatives are strictly monotone in these intervals. This prevents $\overline{h}''_{\Phi_1}(x)$ to be zero and thus we get a contradiction. Hence, there is at most one solution to $\overline{h}'_{\Phi_1}(x)=0$ in $(0,\phi_1)$. The argument for the interval $(\phi_1,1)$ is the same.

A direct computation gives

$$
\begin{aligned}
\overline{h}'_{\Phi_1}(x) &= 1-(L^{0,-1})'(L^{1,-1}(x))(L^{1,-1})'(x)=1-\frac{1}{(L_0{}')(L^{0,-1}(L^{1,-1}(x)))}\frac{1}{(L_1{}')(L^{1,-1}(x))}\\
&= 1-\frac{\int_0^1 w[F^{-1}(L^{0,-1}(L^{1,-1}(x)))]du}{w[F^{-1}(L^{0,-1}(L^{1,-1}(x)))]}\frac{\int_0^1 w[L^{0,-1}(L^{1,-1}(x))]du}{w[L^{0,-1}(L^{1,-1}(x))]}. \qquad (571)
\end{aligned}
$$

Hence we have $\overline{h}'_{\Phi_1}(0)=-\infty$ and $\overline{h}'_{\Phi_1}(1)=-\infty$. We can also notice that around 0.5 there is mass concentration which produces at least one positive value for $h'_{\Phi_1}(x)$. Therefore at $\phi_1^{(1)}$ and $\phi_1^{(2)}$, the function $h'_{\Phi_1}(x)$ changes its sign which determine the behavior $\Phi_n(x)$ and its crossing properties. Namely, as hinted at the beginning, the positioning of the $\Phi_1(x)$ above and below the diagonal is analogous to the situation of $L^{n,-1}(x)$ for any n. The crossing pattern is: $\Phi_1(x)>x$ for $x\in(0,\phi_1)$ and $\Phi_1(x)<x$ for $x\in(\phi_1,1)$, while at each of these intervals the points $\phi_1^{(1)}$ and $\phi_1^{(2)}$ just produce a maximal distance between $\Phi_1(x)$ and the diagonal. Furthermore, the described behavior of $\overline{h}'_{\Phi_1}(x)$ gives that $\overline{h}'_{\Phi_1}(x)>0$, and thus $\Phi'_1(x)<1$, hold for $x\in(\phi_1^{(1)},\phi_1^{(2)})$ and $\overline{h}'_{\Phi_1}(x)<0$, and thus $\Phi'_1(x)>1$, hold for $x\in[0,\phi_1^{(1)})\cup(\phi_1^{(2)},1]$ with the corresponding zeros attained at $\phi_1^{(1)}$ and $\phi_1^{(2)}$. So the interval I_1 is $[\phi_1^{(1)},\phi_1^{(2)}]$.

Now we can move to the case n. From $\Phi_n(x)=L^{0,-1}(...(L^{n-1,-1}(L^{n,-1}(x)))$ we have $\Phi_n(x)=\Phi_{n-1}(L^{n,-1}(x))$. Either inductively, or simply by change of notation, we can notice that everything proved above also holds for $\Phi_n(x)$. This is essentially due to the again appearing bifunctional composite form of $\Phi_n(x)$ which was the case also for $\Phi_1(x)=L^{0,-1}(L^{1,-1}(x))$. Exactly, it allows to make the induction step by assuming for n that $\Phi_n(x)$ obeys the conditions of the claim and then prove them for $n+1$. The proof will verbally reiterate the proof above for $n=1$ with a change of notation to reflect the fact that here we work with Φ_{n-1} instead of $\Phi_0\equiv L^{0,-1}$.

We may also finally notice that if we set $\Psi_n(x)=\Phi_n^{-1}(x)$ then $\Psi_n(x)=L_n(...(L_1(L_0(x)))$. The function $\Psi_n(x)$ resembles the crossing behavior of $L_n(x)$. Namely, $\Psi_n(x)$ has two fixed points at 0 and 1. Additionally, it has a unique fixed point in $(0,1)$. If we denote the latter by ψ_n, then $\Psi_n(x)<x$ holds in the interval $(0,\psi_n)$ and $\Psi_n(x)>x$ holds in the interval $(\psi_n,1)$. There is also a closed interval $I_n\subset(0,1)$ such that $\psi_n\in I_n$ and the derivative satisfies $\Psi'_n(x)\geq 1$ for all $x\in I_n$.

The proof of the preceding lemma also yields the following immediate corollary. □

Corollary 104. *The inequality $\Phi'_n(x)<1$ holds for all x in the interval $(\phi_n^{(1)},\phi_n^{(2)})$.*

We now relate the evolution of the Φ_n to that of the crossing points c_n from *Lemma 60.*

Lemma 105. *For every $n\geq 0$ and $x\in(0,1)$,*

$$sgn(\Phi_{n+1}(x)-\Phi_n(x))=sgn(T_{n+1}(x)-x)=sgn(c_{n+1}-x). \qquad (572)$$

Proof. By definition,

$$\Phi_{n+1}(x)=\Phi_n(T_{n+1}(x)). \qquad (573)$$

Since Φ_n is strictly increasing, the sign of $\Phi_{n+1}(x) - \Phi_n(x)$ is the same as the sign of $T_{n+1}(x) - x$, i.e.

$$sgn(\Phi_{n+1}(x) - \Phi_n(x)) = sgn(T_{n+1}(x) - x). \tag{574}$$

It remains to relate the sign of $T_{n+1}(x) - x$ to the position of x relative to c_{n+1}. Because T_{n+1} is the inverse of L_{n+1} and L_{n+1} is strictly increasing, we have

$$T_{n+1}(x) > x \iff L_{n+1}(T_{n+1}(x)) > L_{n+1}(x) \iff x > L_{n+1}(x). \tag{575}$$

By *Lemma 60*, $L_{n+1}(x) < x$ for $x < c_{n+1}$ and $L_{n+1}(x) > x$ for $x > c_{n+1}$. Thus

$$T_{n+1}(x) > x \iff x < c_{n+1}. \tag{576}$$

Similarly, $T_{n+1}(x) < x$ if and only if $x > c_{n+1}$. Therefore

$$sgn(T_{n+1}(x) - x) = sgn(c_{n+1} - x), \tag{577}$$

which combined with the first identity gives (572). ☐

We may observe that *Lemma* 105 gives that as n grows, the evolution of $\Phi_n(x)$ is tightly controlled by the motion of the c_n's. Next, we relate the latter to the ϕ_n's.

Lemma 106. *For each $n \geq 0$, the fixed point ϕ_{n+1} lies between ϕ_n and c_{n+1}*

$$\min\{\phi_n, c_{n+1}\} \ \leq \ \phi_{n+1} \ \leq \ \max\{\phi_n, c_{n+1}\}. \tag{578}$$

Proof. Define $k_{n+1}(x) = -\overline{h}_{\Phi_n} = \Phi_{n+1}(x) - x$. Then k_{n+1} is continuous, strictly negative on $(\phi_{n+1}, 1)$ and strictly positive on $(0, \phi_{n+1})$, and has a unique zero at $x = \phi_{n+1}$. We evaluate k_{n+1} at ϕ_n and c_{n+1}.

At $x = \phi_n$ we have

$$k_{n+1}(\phi_n) = \Phi_{n+1}(\phi_n) - \phi_n = \Phi_{n+1}(\phi_n) - \Phi_n(\phi_n), \tag{579}$$

so by *Lemma* 105 with $x = \phi_n$,

$$sgn(k_{n+1}(\phi_n)) = sgn(\Phi_{n+1}(\phi_n) - \Phi_n(\phi_n)) = sgn(c_{n+1} - \phi_n). \tag{580}$$

At $x = c_{n+1}$ we get

$$k_{n+1}(c_{n+1}) = \Phi_{n+1}(c_{n+1}) - c_{n+1} = \Phi_n(T_{n+1}(c_{n+1})) - c_{n+1} = \Phi_n(c_{n+1}) - c_{n+1}. \tag{581}$$

By the single-crossing property for Φ_n from *Lemma* 103, we have

$$sgn(k_{n+1}(c_{n+1})) = sgn(\Phi_n(c_{n+1}) - c_{n+1}) = sgn(\phi_n - c_{n+1}). \tag{582}$$

Thus $k_{n+1}(\phi_n)$ and $k_{n+1}(c_{n+1})$ have opposite signs. Since k_{n+1} is continuous and has a unique zero ϕ_{n+1}, it follows that ϕ_{n+1} lies strictly between ϕ_n and c_{n+1}, and this is exactly (578). ☐

Remark 107. *Lemma* 106 formalizes the idea that the new fixed point of the compound map Φ_{n+1} is "squeezed" between the previous compound fixed point ϕ_n and the fresh base fixed point c_{n+1}. Over many iterations, this repeated squeezing forces the sequences $\{\phi_n\}_{n=0}^{\infty}$ and $\{c_n\}_{n=0}^{\infty}$ to track each other closely.

We have even a stronger result, which comes as a corollary of *Lemma* 103.

Corollary 108. *For every $n \geq 0$, the unique interior fixed point $\phi_{n+1} \in (0, 1)$ of Φ_{n+1} satisfies*

$$\min\{\, L_{n+1}(c_n),\, c_{n+1}\,\} \ \leq \ \phi_{n+1} \ \leq \ \max\{\, L_{n+1}(c_n),\, c_{n+1}\,\}. \tag{583}$$

Proof. This is exactly the localization of the unique zero of the deviation function $k_{n+1}(x) = \Phi_{n+1}(x) - x$ that is obtained inside the proof of *Lemma* 103. Indeed, in that proof we analyze the sign contributions in k_{n+1} and shows (after possibly exchanging the roles of the two endpoints) that $k_{n+1}(c_{n+1})$ and $k_{n+1}(L_{n+1}(c_n))$ have opposite signs. Since k_{n+1} is continuous and has a unique zero in $(0, 1)$ (namely ϕ_{n+1}, by *Lemma* 103), the *Intermediate Value Theorem* implies that this zero must lie between the two points c_{n+1} and $L_{n+1}(c_n)$. This yields (583). ☐

Local equicontinuity and subsequential uniform convergence of $\Phi_n(x)$

We begin by establishing a basic contraction principle for increasing self-maps with a single interior fixed point. The proof relies on two key results: a *Fejer-type inequality* (see [16]) and a *1-Lipschitz bound.*

Lemma 109. *Let $f : [0, 1] \to [0, 1]$ be continuous, strictly increasing, with a unique interior fixed point $p \in (0, 1)$ and single-crossing pattern*

$$f(x) > x \quad (x < p), \qquad f(x) < x \quad (x > p). \tag{584}$$

Then:

1. *For every $x \in [0, 1]$*

$$|f(x) - p| \le |x - p|, \tag{585}$$

   with strict inequality if $x \neq p$;

2. *If x, y lie on the same side of p (both $\le p$ or both $\ge p$), then*

$$|f(x) - f(y)| \le |x - y|.$$

Proof. If $x = p$ the inequality is trivial. Suppose $x > p$. Then $f(x) \in [p, x)$, hence $0 \le f(x) - p < x - p$, so $|f(x) - p| < |x - p|$. If $x < p$, then $f(x) \in (x, p]$, so $0 \le p - f(x) < p - x$ and again $|f(x) - p| < |x - p|$. That proves the first part of the lemma.

For the second part, suppose $p \le x < y$; the case $x < y \le p$ is analogous. Then $f(x) \ge p$ and $f(y) \le y$. Also f increasing implies $f(x) \le f(y)$. Thus $0 \le f(y) - f(x) \le y - x$. So

$$|f(y) - f(x)| \le |y - x|. \tag{586}$$

□

We can now establish subsequential convergence of the compound maps Φ_n.

Lemma 110. *For every $n \ge 0$ and all $x, y \in [0, 1]$,*

$$|\Phi_n(x) - \Phi_n(y)| \le |x - y|. \tag{587}$$

In particular, the family $(\Phi_n)_{n \ge 0}$ is equicontinuous and uniformly bounded on $[0, 1]$.

Proof. Fix n. By *Lemma 103*, Φ_n is continuous, strictly increasing, and has a unique interior fixed point ϕ_n with the single-crossing pattern

$$\Phi_n(x) > x \ (x < \phi_n), \qquad \Phi_n(x) < x \ (x > \phi_n). \tag{588}$$

Apply *Lemma 109* with $f = \Phi_n$ and $p = \phi_n$. If x, y lie on the same side of ϕ_n, then *Lemma 109* gives

$$|\Phi_n(x) - \Phi_n(y)| \le |x - y|. \tag{589}$$

If $x < \phi_n < y$, then by triangle inequality and *Lemma 109* on each side,

$$|\Phi_n(x) - \Phi_n(y)| \le |\Phi_n(x) - \Phi_n(\phi_n)| + |\Phi_n(\phi_n) - \Phi_n(y)| \le |x - \phi_n| + |y - \phi_n| = |x - y|. \tag{590}$$

This proves (587). Uniform boundedness is clear since $\Phi_n([0, 1]) \subset [0, 1]$. □

Next, we prove an auxiliary result using the fact that all the crossing point of the subsequential limits of L_n are the same.

Claim 111. *Assume that*

$$c_n \longrightarrow c^* \in (0, 1). \tag{591}$$

Then

$$|L_{n+1}(c_n) - c_{n+1}| \longrightarrow 0 \qquad \text{as } n \to \infty. \tag{592}$$

Proof. Suppose, by contradiction, that the above conclusion is false. Then there exist $\varepsilon_0 > 0$ and a strictly increasing sequence of indices $\{n_k\}_{k\geq 1}$ such that

$$|L_{n_k+1}(c_{n_k}) - c_{n_k+1}| \geq \varepsilon_0 \qquad \text{for all } k. \tag{593}$$

By *Claim 64*, the family $\{L_n\}_{n\geq 0}$ is relatively compact in $C([0,1])$. Hence, after passing to a further subsequence (not relabeled), there exists a continuous increasing function $L^* : [0,1] \to [0,1]$ such that

$$L_{n_k+1} \longrightarrow L^* \quad \text{uniformly on } [0,1]. \tag{594}$$

Since $c_n \to c^*$ by assumption, we also have

$$c_{n_k} \longrightarrow c^*, \qquad c_{n_k+1} \longrightarrow c^*. \tag{595}$$

For each k, the point c_{n_k+1} is a fixed point of L_{n_k+1}, hence

$$L_{n_k+1}(c_{n_k+1}) = c_{n_k+1}. \tag{596}$$

Passing to the limit using (594) yields

$$L^*(c^*) = c^*. \tag{597}$$

Again using uniform convergence and $c_{n_k} \to c^*$, we obtain

$$L_{n_k+1}(c_{n_k}) \longrightarrow L^*(c^*) = c^*. \tag{598}$$

Together with $c_{n_k+1} \to c^*$, this implies

$$|L_{n_k+1}(c_{n_k}) - c_{n_k+1}| \longrightarrow 0, \tag{599}$$

which contradicts (593). Therefore, (592) holds. □

We now use the convergence of c_n and *Lemma 106* to deduce convergence of ϕ_n.

Claim 112. *The cluster set of $\{\phi_n\}_{n\geq 0}$ coincides with the cluster set of $\{c_n\}_{n\geq 0}$. In particular, if $c_n \to c^* \in (0,1)$, then*

$$\phi_n \to \phi^* = c^*. \tag{600}$$

Proof. The sequence $\{\phi_n\}_{n\geq 0}$ is contained in $[0,1]$, hence has at least one cluster point. Let ϕ be an arbitrary cluster point of $\{\phi_n\}_{n\geq 0}^{\infty}$. Then there exists a strictly increasing sequence of indices $\{n_k\}_{k\geq 1}$ such that

$$\phi_{n_k} \to \phi. \tag{601}$$

Since $c_n \to c^*$ by assumption, we also have

$$c_{n_k+1} \to c^*. \tag{602}$$

Next, from *Corollary 108,* we have

$$|\phi_{n+1} - c_{n+1}| \ \leq \ |L_{n+1}(c_n) - c_{n+1}| \tag{603}$$

and thus also

$$|\phi_{n_k+1} - c_{n_k+1}| \ \leq \ |L_{n_k+1}(c_{n_k}) - c_{n_k+1}| \ . \tag{604}$$

Passing to the limit along $k \to \infty$ and using (592) from *Claim 111* yields $\phi = c^*$. Consequently, the sequence $\{\phi_n\}_{n=0}^{\infty}$ cannot oscillate between distinct cluster points; it must converge to the unique point c^*. Thus, the cluster set is the singleton $\{c^*\}$, and $\phi_n \longrightarrow c^*$. □

Next, we establish the pointwise convergence of $\Phi_n(x)$ using a monotonicity argument that relies on *Lemma* 105 and *Lemma* 82.

Lemma 113. *For every fixed $x \in (0,1)$, the sequence $\{\Phi_n(x)\}_{n=0}^{\infty}$ converges pointwise*

$$\lim_{n\to\infty} \Phi_n(x) = \Phi_\infty(x), \tag{605}$$

where we denoted the limit by $\Phi_\infty(x)$.

Proof. We consider two cases.

Case 1: $x \neq c^*$. By *Lemma* 105,

$$sgn(\Phi_{n+1}(x) - \Phi_n(x)) = sgn(c_{n+1} - x). \tag{606}$$

Since by *Lemma* 82 $c_n \to c^*$, with $c^* = \frac{1}{2}$, there exists N such that for all $n \geq N$,

$$\operatorname{sgn}(c_{n+1} - x) = \operatorname{sgn}(c^* - x),$$

i.e., the sign of $c_{n+1} - x$ does not change for $n \geq N$. For n sufficiently large it is constant: positive if $x < c^*$ and negative if $x > c^*$. Thus, for large n, the sequence $\{\Phi_n(x)\}_{n=0}^{\infty}$ is monotone. Being bounded in $[0,1]$, it must converge. Let

$$\lim_{n\to\infty} \Phi_n(x) = \Phi_\infty(x). \tag{607}$$

We still need to identify $\Phi_\infty(x)$.

Case 2: $x = c^*$. Applying *Lemma* 109 to $f = \Phi_n$ and $p = \phi_n$ with $x = \phi^* = c^*$ yields

$$|\Phi_n(\phi^*) - \phi_n| \leq |\phi^* - \phi_n|. \tag{608}$$

Hence

$$|\Phi_n(c^*) - c^*| = |\Phi_n(\phi^*) - \phi^*| \leq |\Phi_n(\phi^*) - \phi_n| + |\phi_n - \phi^*| \leq 2|\phi_n - \phi^*|. \tag{609}$$

Since $\phi_n \to \phi^*$, the right-hand side tends to 0, and thus $\Phi_n(c^*) \to c^*$.

Combining the two cases shows that for every $x \in (0,1)$ the limit $\lim_{n\to\infty} \Phi_n(x)$ exists. □

Remark 114. Far from c^*, the evolution of $\Phi_n(x)$ is essentially monotone: once the crossing points c_n have moved past x in one direction, the increments $\Phi_{n+1}(x) - \Phi_n(x)$ keep the same sign. At $x = \phi^*$, the fixed points ϕ_n act as anchors: each Φ_n pulls x towards ϕ_n, and as $\phi_n \to \phi^*$, this pull becomes more and more concentrated, forcing $\Phi_n(\phi^*)$ to converge to ϕ^* as well.

Remark 114 leads us to conjecture a stronger result: (i) the convergence of $\Phi_n(x)$ to $\Phi_\infty(x)$ is uniform, (ii) $c^* = \phi^* = \frac{1}{2}$, (iii) $\Phi_\infty(x)$ is the constant ϕ^*.

We now present a general argument for subsequential convergence that does not rely on the previously established pointwise convergence, after which we will establish the connection between these two modes of convergence within the context of our problem.

Lemma 115. *Let $\{n_k\}_{k\geq 0}$ be any strictly increasing sequence with $n_k \to \infty$. Then there exists a further subsequence (still denoted $\{n_k\}_{k\geq 0}$) and a map*

$$\Phi^* : (0,1) \to (0,1) \tag{610}$$

such that for every $0 < \delta < \frac{1}{2}$,

$$\sup_{x\in[\delta,\,1-\delta]} \left|\Phi_{n_k}(x) - \Phi^*(x)\right| \longrightarrow 0 \qquad (k \to \infty). \tag{611}$$

In particular, Φ^* is continuous and increasing on $(0,1)$.

Moreover, the fixed points $\{\phi_{n_k}\}_{k\geq 0}$ have a further subsequence (still denoted $\{\phi_{n_k}\}_{k\geq 0}$) converging to some $\phi^* \in [0,1]$. If in addition $\phi^* \in (0,1)$, then

$$\Phi^*(\phi^*) = \phi^*. \tag{612}$$

Proof. By *Lemma 110*, each $\Phi_n : [0,1] \to [0,1]$ is 1–*Lipschitz*, hence the family $\{\Phi_n\}_{n\ge 0}$ is equicontinuous and uniformly bounded on $[0,1]$.

Step 1: Uniform convergence on one compact subinterval. Fix $0 < \delta < \frac{1}{2}$ and set $I_\delta = [\delta, 1-\delta]$. The restriction family $\{\Phi_{n_k}|_{I_\delta}\}_{k\ge 0}$ is equicontinuous and uniformly bounded on the compact interval I_δ. By *Arzelà–Ascoli*, there exists a subsequence of indices $\{n_k^{(\delta)}\}_{k\ge 0}$ and a continuous function $\Phi^{(\delta)} : I_\delta \to [0,1]$ such that

$$\sup_{x\in I_\delta} \left|\Phi_{n_k^{(\delta)}}(x) - \Phi^{(\delta)}(x)\right| \to 0 \qquad (k\to\infty). \tag{613}$$

Since each Φ_n is increasing, the uniform limit $\Phi^{(\delta)}$ is also increasing.

Step 2: Choosing a nested exhaustion of $(0,1)$. Define

$$\delta_m = 2^{-m} \qquad (m\ge 2), \qquad I_m = I_{\delta_m} = [\delta_m,\, 1-\delta_m]. \tag{614}$$

Then I_m are compact intervals with

$$I_2 \subset I_3 \subset I_4 \subset \cdots, \qquad \bigcup_{m\ge 2} I_m = (0,1). \tag{615}$$

(Indeed, for any $x\in(0,1)$ choose m large so that $\delta_m < x < 1-\delta_m$.)

Step 3: Iterated extractions. We now build a chain of subsequences by induction on m.

- Start with the original subsequence $\{n_k^{(1)}\}_{k\ge 0} = \{n_k\}_{k\ge 0}$;
- For $m = 2$: apply Step 1 with $\delta = \delta_2$ to the sequence $\{\Phi_{n_k^{(1)}}\}_{k\ge 0}$ on I_2. Obtain a subsequence $\{n_k^{(2)}\}_{k\ge 0} \subset \{n_k^{(1)}\}_{k\ge 0}$ and a continuous increasing limit $\Phi^{(2)} : I_2 \to [0,1]$ such that

$$\sup_{x\in I_2} \left|\Phi_{n_k^{(2)}}(x) - \Phi^{(2)}(x)\right| \to 0; \tag{616}$$

- For $m = 3$: apply Step 1 with $\delta = \delta_3$ to the sequence $\{\Phi_{n_k^{(2)}}\}_{k\ge 0}$ on I_3. Obtain a subsequence $\{n_k^{(3)}\}_{k\ge 0} \subset \{n_k^{(2)}\}_{k\ge 0}$ and a continuous increasing limit $\Phi^{(3)} : I_3 \to [0,1]$ such that

$$\sup_{x\in I_3} \left|\Phi_{n_k^{(3)}}(x) - \Phi^{(3)}(x)\right| \to 0; \tag{617}$$

- Continue inductively: having constructed $\{n_k^{(m-1)}\}_{k\ge 0}$, apply Step 1 with $\delta = \delta_m$ on I_m to obtain a further subsequence $\{n_k^{(m)}\}_{k\ge 0} \subset \{n_k^{(m-1)}\}_{k\ge 0}$ and a continuous increasing limit $\Phi^{(m)} : I_m \to [0,1]$ such that

$$\sup_{x\in I_m} \left|\Phi_{n_k^{(m)}}(x) - \Phi^{(m)}(x)\right| \to 0. \tag{618}$$

Thus we have nested subsequences

$$\{n_k^{(2)}\} \supset \{n_k^{(3)}\} \supset \cdots \tag{619}$$

and uniform convergence on I_m along $\{n_k^{(m)}\}$.

Step 4: The diagonal subsequence (Cantor Diagonalization).

The subsequences constructed above are chosen nested, that is,

$$\{n_k^{(m+1)}\}_{k\ge 0} \subset \{n_k^{(m)}\}_{k\ge 0} \qquad \text{for every } m\ge 2. \tag{620}$$

After discarding finitely many terms if necessary, define the diagonal subsequence $\{n_k^{\mathrm{diag}}\}_{k\ge 0}$ by

$$n_k^{\mathrm{diag}} = n_k^{(k+2)}, \qquad k\ge 0. \tag{621}$$

Fix $m\ge 2$. If $k\ge m-2$, then $k+2\ge m$, and by the nestedness of the subsequences we have

$$n_k^{\mathrm{diag}} = n_k^{(k+2)} \in \{n_j^{(k+2)} : j\ge 0\} \subset \{n_j^{(m)} : j\ge 0\}. \tag{622}$$

Thus, for every fixed $m \geq 2$, the tail of the diagonal subsequence $\{n_k^{\mathrm{diag}}\}_{k\geq 0}$ is a subsequence of $\{n_k^{(m)}\}_{k\geq 0}$. Therefore, using (618), we obtain

$$\sup_{x\in I_m} \left|\Phi_{n_k^{\mathrm{diag}}}(x) - \Phi^{(m)}(x)\right| \longrightarrow 0 \qquad (k\to\infty). \tag{623}$$

Consequently, the diagonal subsequence converges uniformly on each compact interval I_m. Since the intervals I_m exhaust $(0,1)$, this gives locally uniform convergence on $(0,1)$.

Step 5: Consistency of the limits on overlaps. Fix $m < \ell$. Then $I_m \subset I_\ell$. We claim that $\Phi^{(\ell)} = \Phi^{(m)}$ on I_m. Indeed, along the diagonal subsequence we have by (618)

$$\Phi_{n_k^{\mathrm{diag}}} \to \Phi^{(m)} \text{ uniformly on } I_m, \qquad \Phi_{n_k^{\mathrm{diag}}} \to \Phi^{(\ell)} \text{ uniformly on } I_\ell, \tag{624}$$

hence also uniformly on $I_m \subset I_\ell$. Uniform limits on the same set are unique, so $\Phi^{(\ell)} = \Phi^{(m)}$ on I_m.

Step 6: Definition of Φ^* and proof of (611). By Step 5, the family $\{\Phi^{(m)}\}_{m\geq 2}$ is compatible on overlaps. Define $\Phi^* : (0,1)\to(0,1)$ by

$$\Phi^*(x) = \Phi^{(m)}(x) \quad \text{whenever } x\in I_m. \tag{625}$$

This is well-defined: if $x\in I_m\cap I_\ell$, Step 5 gives $\Phi^{(m)}(x) = \Phi^{(\ell)}(x)$.

Now fix an arbitrary $0<\delta<\frac{1}{2}$. Choose $m\geq 2$ such that $\delta_m \leq \delta$. Then $[\delta, 1-\delta]\subset I_m$, and by (623),

$$\sup_{x\in[\delta,1-\delta]} \left|\Phi_{n_k^{\mathrm{diag}}}(x) - \Phi^*(x)\right| \leq \sup_{x\in I_m} \left|\Phi_{n_k^{\mathrm{diag}}}(x) - \Phi^{(m)}(x)\right| \to 0, \tag{626}$$

which is exactly (611) (after relabelling n_k^{diag} as n_k).

Continuity and monotonicity of Φ^* on $(0,1)$ now follow because on each I_m the function Φ^* agrees with the continuous increasing function $\Phi^{(m)}$.

Step 7: Subsequence convergence of fixed points and the conditional fixed-point identity. Since each $\phi_{n_k}\in(0,1)$, the sequence $\{\phi_{n_k}\}$ is bounded in $[0,1]$, hence has a convergent sub-subsequence (still denoted $\{\phi_{n_k}\}$) with $\phi_{n_k}\to\phi^*\in[0,1]$.

Assume now that $\phi^*\in(0,1)$. Choose $\delta>0$ such that $\phi^*\in[\delta,1-\delta]$. Then for all large k, $\phi_{n_k}\in[\delta,1-\delta]$ as well. By (611), we have uniform convergence $\Phi_{n_k}\to\Phi^*$ on $[\delta,1-\delta]$. Therefore,

$$\left|\Phi_{n_k}(\phi_{n_k}) - \Phi^*(\phi^*)\right| \leq \left|\Phi_{n_k}(\phi_{n_k}) - \Phi^*(\phi_{n_k})\right| + \left|\Phi^*(\phi_{n_k}) - \Phi^*(\phi^*)\right| \to 0, \tag{627}$$

using uniform convergence for the first term and continuity of Φ^* for the second. Since $\Phi_{n_k}(\phi_{n_k}) = \phi_{n_k}$ for all k, the left-hand side also equals $|\phi_{n_k} - \Phi^*(\phi^*)|\to 0$, hence $\Phi^*(\phi^*) = \phi^*$, which is (612).

We must note that the *Cantor diagonalization* was needed because uniform convergence on one compact interval does not automatically give a single subsequence that converges uniformly on all compact intervals, and the diagonal argument is exactly what makes one subsequence work on the whole exhaustion of $(0,1)$. □

Remark 116. For every n we have $\Phi_n(0) = 0$ and $\Phi_n(1) = 1$. We later prove that $\Phi_n(x)\to\frac{1}{2}$ for each $x\in(0,1)$. Thus any Φ^* limit map equals $\frac{1}{2}$ on $(0,1)$ but has endpoint values $0,1$, hence it is discontinuous at 0 and 1. Therefore uniform convergence on $[0,1]$ is impossible, and the correct compactness notion is uniform convergence on compact subsets of $(0,1)$ (i.e. local uniform convergence).

By combining *Lemma 115* and *Lemma 113*, we obtain the following corollary.

Corollary 117. *The sequence $\{\Phi_n\}_{n\geq 0}$ converges to Φ_∞ locally uniformly on $(0,1)$. Moreover, the fixed points ϕ_n converge to ϕ^*. If $\phi^*\in(0,1)$, then*

$$\Phi_\infty(\phi^*) = \phi^*. \tag{628}$$

Proof. Fix any compact interval $[\delta,1-\delta]\subset(0,1)$. By *Lemma 110*, the family $\{\Phi_n\}_{n\geq 0}$ is equicontinuous and uniformly bounded on $[\delta,1-\delta]$.

Let $\{n_k\}_{k\geq 0}$ be any sequence with $n_k \to \infty$. By *Lemma 115*, there exists a subsequence (still denoted n_k) and a continuous increasing function Φ^* such that

$$\sup_{x\in[\delta,1-\delta]} |\Phi_{n_k}(x) - \Phi^*(x)| \to 0. \tag{629}$$

On the other hand, by *Lemma 113*, the full sequence $\Phi_n(x)$ converges pointwise to $\Phi_\infty(x)$ for every $x \in (0,1)$. Since uniform convergence implies pointwise convergence, we must have

$$\Phi^*(x) = \Phi_\infty(x) \quad \text{for all } x \in [\delta, 1-\delta]. \tag{630}$$

Since the choice of subsequence was arbitrary, every subsequence admits a further subsequence converging uniformly on $[\delta, 1-\delta]$ to the same limit Φ_∞. This implies that the full sequence Φ_n converges uniformly on $[\delta, 1-\delta]$. As $\delta > 0$ was arbitrary, the convergence is locally uniform on $(0,1)$.

The convergence $\phi_n \to \phi^*$ was established in *Claim 112*. If $\phi^* \in (0,1)$, then by continuity of Φ_∞,

$$\Phi_\infty(\phi^*) = \lim_{n\to\infty} \Phi_n(\phi_n) = \lim_{n\to\infty} \phi_n = \phi^*. \tag{631}$$

□

Convergence to a constant limit of Φ_n

Our next goal is to further characterize the limits of the sequences $\{\Phi_n\}_{n=0}^{\infty}$ and $\{\phi_n\}_{n\geq 0}^{\infty}$.

Lemma 118. *Let $[\delta, \eta] \subset (0,1)$ be fixed. Then for any sequence of indices $\{n_k\}_{k=0}^{\infty}$ with $n_k \to \infty$, there exists a subsequence (still denoted n_k) and a limit inverse map T^* such that*

$$\Phi_\infty(x) = \Phi_\infty(T^*(x)) \quad \textit{for all } x \in [\delta, \eta]. \tag{632}$$

Proof. By *Claim 64*, from the sequence $\{n_k\}$ we may extract a subsequence (not relabeled) such that

$$L_{n_k} \to L^* \quad \text{uniformly on } [0,1], \tag{633}$$

and therefore

$$T_{n_k} = L_{n_k}^{-1} \to T^* = (L^*)^{-1} \quad \text{uniformly on } [0,1]. \tag{634}$$

For each k and $x \in [\delta, \eta]$ we have the exact identity

$$\Phi_{n_k}(x) = \Phi_{n_k-1}(T_{n_k}(x)). \tag{635}$$

Since T^* is continuous and maps $(0,1)$ into itself, the image $T^*([\delta, \eta])$ is a compact subset of $(0,1)$. Hence there exist numbers $0 < \delta' < \eta' < 1$ such that

$$T^*([\delta, \eta]) \subset (\delta', \eta'). \tag{636}$$

By uniform convergence $T_{n_k} \to T^*$ on $[0,1]$, there exists K such that for all $k \geq K$,

$$T_{n_k}(x) \in [\delta', \eta'] \quad \text{for all } x \in [\delta, \eta]. \tag{637}$$

By *Corollary 117*, $\Phi_n \to \Phi_\infty$ uniformly on $[\delta', \eta']$. Therefore, for $x \in [\delta, \eta]$,

$$\Phi_{n_k-1}(T_{n_k}(x)) \to \Phi_\infty(T^*(x)), \qquad \Phi_{n_k}(x) \to \Phi_\infty(x). \tag{638}$$

Passing to the limit in the identity (635) yields

$$\Phi_\infty(x) = \Phi_\infty(T^*(x)), \qquad x \in [\delta, \eta]. \tag{639}$$

which proves the claim. □

We next consider the convergence of orbits under a cluster inverse T^* and then use it to show that Φ_∞ must be constant.

Lemma 119. *Let T^* be a cluster inverse. Then T^* has a unique interior fixed point c^*, and for any $x \in (0,1)$,*

$$T^{*\circ m}(x) \longrightarrow c^* \quad \text{as } m \to \infty. \tag{640}$$

Proof. By *Corollary* 73, the limit map L^* has a unique interior fixed point c^* with $L^*(x) < x$ for $x < c^*$ and $L^*(x) > x$ for $x > c^*$. Inverting, $T^*(x) > x$ for $x < c^*$ and $T^*(x) < x$ for $x > c^*$. Thus the orbit $\{T^{*\circ m}(x)\}_{m=0}^{\infty}$, i.e. $T^*(\overbrace{\ldots}^{m-2}(T^*(x)))$ with $m \geq 0$, is monotone and bounded for each x, hence convergent (see [12, Chapter 3.2] for details on orbits' definition), and the limit must be the unique fixed point c^*.

Moreover, by the *Mean Value Theorem* combined with *Lemma* 60 and *Corollary 61*, there is a neighborhood U of c^* and $q \in (0,1)$ such that

$$|T^*(y) - c^*| \leq q|y - c^*| \quad \text{for all } y \in U.$$

For any $x \in (0,1)$, monotonicity of T^* and its fixed-point structure imply (see [12, Chapter 5.2] for details) that $T^{*\circ m}(x)$ eventually enters U (if $x < c^*$, the orbit increases into U; if $x > c^*$, it decreases into U). Once inside U, the contraction estimate ensures exponential convergence to c^*. Thus $T^{*\circ m}(x) \to c^*$ for all $x \in (0,1)$. □

Lemma 120. *The limit map Φ_∞ is constant on $(0,1)$ and equal to c^**

$$\Phi_\infty(x) \equiv c^* = \frac{1}{2}, \quad x \in (0,1). \tag{641}$$

Proof. Fix an arbitrary compact interval $[\delta, \eta] \subset (0,1)$. By *Lemma* 117, $\Phi_n \to \Phi_\infty$ uniformly on $[\delta, \eta]$.

Let $\{n_k\}_{k=0}^{\infty}$ be any sequence of indices with $n_k \to \infty$. By *Claim* 64 we may extract a subsequence (still denoted n_k) such that $L_{n_k} \to L^*$ and $T_{n_k} \to T^*$ uniformly, with T^* the inverse of a limit L^* satisfying (267). By *Lemma* 118,

$$\Phi_\infty(x) = \Phi_\infty(T^*(x)), \quad x \in [\delta, \eta]. \tag{642}$$

Iterating this relation, for all $m \in N$,

$$\Phi_\infty(x) = \Phi_\infty(T^{*\circ m}(x)), \quad x \in [\delta, \eta]. \tag{643}$$

By *Lemma* 119, $T^{*\circ m}(x) \to c^*$ for every $x \in [\delta, \eta]$. By continuity of Φ_∞ (*Corollary 117*),

$$\Phi_\infty(x) = \lim_{m\to\infty} \Phi_\infty(T^{*\circ m}(x)) = \Phi_\infty(c^*) \tag{644}$$

for all $x \in [\delta, \eta]$. Thus Φ_∞ is constant on $[\delta, \eta]$, with value $c = \Phi_\infty(c^*)$.

Since $[\delta, \eta] \subset (0,1)$ was arbitrary, we conclude that $\Phi_\infty(x) \equiv c$ on all of $(0,1)$. Finally, evaluating at $x = \phi^*$ and using *Lemma* 113 (which gives $\Phi_n(\phi^*) \to \phi^*$), we obtain

$$c = \Phi_\infty(\phi^*) = \lim_{n\to\infty} \Phi_n(\phi^*) = \phi^*. \tag{645}$$

Hence $\Phi_\infty(x) \equiv \phi^*$ for all $x \in (0,1)$. □

Appendix C.1.2

In the previous appendix, we extensively analyzed the function $L_n(x)$ generated by the *Fréchet-Hoeffding lower-bound* iteration (222). We established several of its key properties and characterized its asymptotic behavior. Specifically, we proved the subsequential uniform convergence of the sequence $\{L_n(x)\}_{n\geq 0}$ on $[0,1]$, as well as the convergence of its crossing points, c_n, to $\frac{1}{2}$. We also considered the corresponding sequence of its iterated inverses, $\{\Phi_n(x)\}_{n\geq 0}$, proving the convergence of their crossing points, ϕ_n, to $\frac{1}{2}$ and, more generally, the locally uniform convergence of the sequence itself on $(0,1)$ to $\frac{1}{2}$.

These results provide important insights into the mechanics of the iterative equation (5), particularly under conditions of extreme negative dependence. Furthermore, many of the demonstrated techniques will prove useful for analyzing the general RR_2 case and for making relevant comparisons. Nevertheless, significant questions remain regarding the full convergence of $\{L_n(x)\}_{n\geq 0}$ and the explicit form of its subsequential or full limits.

In this appendix we complete the convergence analysis of the sequence $\{L_n\}_{n\geq 0}$. We assume throughout all the structural facts established earlier in the paper, in particular those from *Appendix C.1.1* concerning monotonicity, convexity/concavity on the relevant subintervals, and the already proved symmetry-defect convergence

$$\|L_n(x) + L_n(1-x) - 1\|_{L^\infty([0,1])} \longrightarrow 0. \tag{646}$$

We also assume relative compactness of $\{L_n\}_{n\geq 0}$ in $C([0,1])$.

The proof is organized as follows.

Outline:

1. We first reduce the problem from the original sequence $\{L_n\}_{n\geq 0}$ to a one-sided profile $\{R_n\}_{n\geq 0}$ adapted to symmetry. As every cluster point of $\{L_n\}_{n\geq 0}$ is symmetric, it becomes uniquely encoded by its reduced profile.
2. We then pass from $\{R_n\}_{n\geq 0}$ to an autonomous dynamical system on a new sequence $\{A_n\}_{n\geq 0}$ and compute the exact *Fréchet derivative* of the corresponding nonlinear map T.
3. The derivative formula yields an explicit pointwise coefficient
$$K_n(x) = \frac{(1-\tau_n(x))p_n(x) + \tau_n(x)(1-p_n(x))}{M_n}, \tag{647}$$
whose strict inequality $K_n(x) < 1$ is the exact contraction criterion.
4. The inequality $K_n(x) < 1$ is proved by a complete branch analysis according to whether $p_n(x) \gtrless \frac{1}{2}$.
5. Finally, we present two ways of closing the argument.
   (i) a direct orbitwise contraction proof, based on the tail coefficients $q_n = \sup_x K_n(x)$;
   (ii) a rigorous compactness/cluster-set proof, which is the argument we regard as conceptually definitive.

The reduced profile R_n and why it is the right variable

Define
$$\gamma(x) = 1 - \sqrt{1-x}, \qquad \eta(x) = 2x - x^2 = \gamma^{-1}(x), \qquad x \in [0,1]. \tag{648}$$

Following the discussion in *Appendix C.1.1*, we define the reduced profile of L_n by

$$R_n(x) = 2L_n\left(\frac{1-\sqrt{1-x}}{2}\right), \qquad x \in [0,1]. \tag{649}$$

The map
$$x \longmapsto \alpha(x) = \frac{1-\sqrt{1-x}}{2} \tag{650}$$

sends $[0,1]$ onto $[0,\frac{1}{2}]$. Thus R_n is exactly the left half of L_n, reparametrized to the whole unit interval and rescaled by the factor 2. This is a useful variable because the symmetry relation

$$L(x) = 1 - L(1-x) \tag{651}$$

determines the whole function from its left half. In particular, if L is symmetric and R is defined by (649), then L can be reconstructed from R by

$$L(x) = \begin{cases} \dfrac{1}{2}\, R(4x(1-x)), & 0 \leq x \leq \dfrac{1}{2}, \\ 1 - \dfrac{1}{2}\, R(4x(1-x)), & \dfrac{1}{2} \leq x \leq 1. \end{cases} \tag{652}$$

Indeed, if $x \in [0, 1/2]$ then $4x(1-x) \in [0,1]$ and

$$\frac{1-\sqrt{1-4x(1-x)}}{2} = x, \tag{653}$$

hence $R(4x(1-x)) = 2L(x)$. For $x \in [1/2, 1]$, symmetry yields

$$L(x) = 1 - L(1-x) = 1 - \frac{1}{2}R(4x(1-x)). \tag{654}$$

Thus, once the limit is known to be symmetric, convergence of R_n is equivalent to convergence of L_n.

The reason this reduction is valid in the present problem is that the symmetry defect has already been proved to vanish uniformly. Therefore every subsequential limit of (L_n) is symmetric. This fact is precisely what allows we to focus on the reduced profiles R_n without losing information.

Lemma 121. *Assume that $R_n \to R_\infty$ uniformly on $[0,1]$ and that (646) holds. Then $L_n \to L_\infty$ uniformly on $[0,1]$, where L_∞ is the symmetric function determined by (652) with $R = R_\infty$.*

Proof. For $x \in [0, \frac{1}{2}]$ we have exactly

$$L_n(x) = \frac{1}{2} R_n(4x(1-x)). \tag{655}$$

Hence

$$\sup_{0 \le x \le 1/2} |L_n(x) - L_\infty(x)| \le \frac{1}{2}\|R_n - R_\infty\|_\infty \longrightarrow 0. \tag{656}$$

Now let $x \in [\frac{1}{2}, 1]$. Set

$$\varepsilon_n(x) = L_n(x) + L_n(1-x) - 1. \tag{657}$$

Then

$$L_n(x) = 1 - L_n(1-x) + \varepsilon_n(x) = 1 - \frac{1}{2}R_n(4x(1-x)) + \varepsilon_n(x), \tag{658}$$

because $1 - x \in [0, \frac{1}{2}]$. Since L_∞ is symmetric,

$$L_\infty(x) = 1 - \frac{1}{2}R_\infty(4x(1-x)). \tag{659}$$

Therefore

$$|L_n(x) - L_\infty(x)| \le \frac{1}{2}\|R_n - R_\infty\|_\infty + \|\varepsilon_n\|_\infty. \tag{660}$$

Taking the supremum over $x \in [\frac{1}{2}, 1]$ and using (646) gives

$$\sup_{1/2 \le x \le 1} |L_n(x) - L_\infty(x)| \le \frac{1}{2}\|R_n - R_\infty\|_\infty + \|L_n(\cdot) + L_n(1-\cdot) - 1\|_\infty \to 0. \tag{661}$$

Combining the two half-interval estimates yields the desired uniform convergence. □

The autonomous map on A_n

The sequence $\{R_n\}_{n \ge 0}$ is not autonomous in its most immediate form. The correct autonomous variable is obtained by integrating the inverse profile.

Define

$$A_n(x) = \frac{\int_0^x R_n^{-1}(s)\,ds}{\int_0^1 R_n^{-1}(s)\,ds}, \qquad x \in [0,1]. \tag{662}$$

Then, by the exact identity established in the discussion above,

$$R_{n+1}(x) = A_n(\gamma(x)), \qquad x \in [0,1]. \tag{663}$$

Equivalently,

$$A_n(x) = R_{n+1}(\eta(x)), \qquad x \in [0,1]. \tag{664}$$

Now define the nonlinear map

$$(TA)(x) = \frac{\int_0^x \eta(A^{-1}(s))\,ds}{\int_0^1 \eta(A^{-1}(s))\,ds}, \qquad x \in [0,1]. \tag{665}$$

Then from (663) and (664) we obtain

$$A_{n+1} = T(A_n). \tag{666}$$

Thus the convergence problem has been reduced to the iteration of the autonomous map T.

For later use, introduce

$$q_n = A_n^{-1}, \qquad p_n(x) = \eta(q_n(x)), \qquad M_n = \int_0^1 p_n(s)\,ds, \qquad \tau_n(x) = A_{n+1}(x). \tag{667}$$

By construction,

$$p_n(x) = R_{n+1}^{-1}(x), \qquad \tau_n(x) = R_{n+2}(\eta(x)). \tag{668}$$

Admissible profiles and basic properties

Let $\mathcal{A}$ denote the class of all functions $A \in C([0,1])$ such that

1. $A(0) = 0$, $A(1) = 1$;
2. A is strictly increasing;
3. A is convex;
4. $A(x) \le x$ on $[0,1]$.

All iterates A_n belong to $\mathcal{A}$ by the results already established for L_n and R_n in *Appendix C.1.1*.

If $A \in \mathcal{A}$, define

$$q = A^{-1}, \qquad p(x) = \eta(q(x)), \qquad M = \int_0^1 p(s)\,ds, \qquad \tau = TA. \tag{669}$$

Then:

1. q is increasing and concave;
2. since $A(x) \le x$, one has $q(x) \ge x$;
3. since $\eta(t) = 2t - t^2$ is increasing and concave on $[0,1]$, the composition $p = \eta \circ q$ is increasing and concave;
4. $p(0) = 0$, $p(1) = 1$;
5. $p(x) \ge \eta(x) = 2x - x^2$ on $[0,1]$, and therefore

$$M = \int_0^1 p(s)\,ds \ge \int_0^1 (2s - s^2)\,ds = \frac{2}{3}. \tag{670}$$

We shall use (670) systematically in what follows.

Fréchet derivative of T

This is the common conceptual core of both proofs. Let $A \in \mathcal{A}$, and define q, p, M, τ as above. Then

$$\tau(x) = \frac{\int_0^x p(s)\,ds}{M}. \tag{671}$$

Differentiating (665) with respect to A yields the exact *Fréchet derivative*.

Lemma 122. *Let $A \in \mathcal{A}$ and $h \in C([0,1])$. Then*

$$DT_A[h](x) = \frac{-\int_0^{q(x)} (2-2t)\,h(t)\,dt + \tau(x)\int_0^1 (2-2t)\,h(t)\,dt}{M}. \tag{672}$$

Consequently,

$$\|DT_A\|_{\infty\to\infty} \le \sup_{x\in[0,1]} \frac{(1-\tau(x))p(x)+\tau(x)(1-p(x))}{M}. \tag{673}$$

For the indexed orbit this becomes

$$K_n(x) = \frac{(1-\tau_n(x))p_n(x)+\tau_n(x)(1-p_n(x))}{M_n}, \qquad \|DT_{A_n}\|_{\infty\to\infty} \le \sup_{x\in[0,1]} K_n(x). \tag{674}$$

Proof. Let $A_\varepsilon = A + \varepsilon h$ and $q_\varepsilon = A_\varepsilon^{-1}$. The standard differentiation formula for inverses gives

$$\left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} q_\varepsilon(x) = -h(q(x)). \tag{675}$$

Since $p_\varepsilon(x) = \eta(q_\varepsilon(x))$, we obtain

$$\left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} p_\varepsilon(x) = (2-2q(x)) \left.\frac{d}{d\varepsilon}\right|_{\varepsilon=0} q_\varepsilon(x) = -(2-2q(x))h(q(x)). \tag{676}$$

Writing (665) as

$$T(A)(x) = \frac{N_A(x)}{D_A}, \qquad N_A(x) = \int_0^x p(s)\,ds, \qquad D_A = \int_0^1 p(s)\,ds = M, \tag{677}$$

and differentiating the quotient yields

$$DT_A[h](x) = \frac{\delta N_A[h](x)}{M} - \frac{N_A(x)}{M^2}\delta D_A[h]. \tag{678}$$

Since q is increasing and $p = \eta \circ q$, the change of variables $s = A(t)$ gives

$$\delta N_A[h](x) = -\int_0^{q(x)} (2-2t)h(t)\,dt, \qquad \delta D_A[h] = -\int_0^1 (2-2t)h(t)\,dt. \tag{679}$$

Substituting this into the quotient rule yields (672).

To derive (673), fix $x \in [0,1]$. Then from (672),

$$|DT_A[h](x)| \le \frac{1}{M}\left(\int_0^{q(x)} (2-2t)|h(t)|\,dt + \tau(x)\int_{q(x)}^1 (2-2t)|h(t)|\,dt\right). \tag{680}$$

Taking $\|h\|_\infty \le 1$ and using

$$\int_0^{q(x)} (2-2t)\,dt = \eta(q(x)) = p(x), \qquad \int_{q(x)}^1 (2-2t)\,dt = 1-p(x), \tag{681}$$

we obtain

$$|DT_A[h](x)| \le \frac{(1-\tau(x))p(x)+\tau(x)(1-p(x))}{M}. \tag{682}$$

Taking the supremum over x proves (673). The indexed formula (674) is just the specialization $A = A_n$. □

Remark 123. The use of the *Fréchet derivative* in the sequel relies on standard *Banach-space* calculus and fixed-point theory. In particular, we use the *Banach-space mean-value theorem* and the contraction principles in the form of [6, Theorem 9.9 (mean value theorem), Theorem 9.27 (contraction principle), Theorem 9.29 (uniform contraction principle)]. The exact derivative formula (672) is the bridge between the nonlinear dynamics of T and the contraction mechanism.

Branch criterion

By (674), to prove strict contractivity it is enough to show

$$K_n(x) < 1. \tag{683}$$

Equivalently,

$$(1-\tau_n(x))p_n(x) + \tau_n(x)(1-p_n(x)) < M_n. \tag{684}$$

Writing $p = p_n(x)$, $\tau = \tau_n(x)$, $M = M_n$, this becomes

$$p + \tau(1-2p) < M. \tag{685}$$

Therefore:

1. if $p > \frac{1}{2}$, then (684) is equivalent to

$$\tau > \frac{p-M}{2p-1}; \tag{686}$$

2. if $p < \frac{1}{2}$, then (684) is equivalent to

$$\tau < \frac{M-p}{1-2p}. \tag{687}$$

We now prove (686) and (687).

The branch $p_n(x) > \frac{1}{2}$

Fix n and $x \in [0,1]$, and abbreviate

$$p = p_n(x), \qquad M = M_n, \qquad \tau = \tau_n(x). \tag{688}$$

We assume

$$p > \frac{1}{2}. \tag{689}$$

We prove (686).

Lemma 124. *For $0 \le s \le x$ we have*

$$p_n(s) \ge \frac{s}{x}\, p. \tag{690}$$

Consequently,

$$\int_0^x p_n(s)\, ds \ge \frac{xp}{2}. \tag{691}$$

Proof. Since p_n is concave and $p_n(0) = 0$, for every $s \in [0,x]$ we can write

$$s = \frac{s}{x}x + \left(1 - \frac{s}{x}\right)0.$$

Concavity gives

$$p_n(s) \ge \frac{s}{x}p_n(x) + \left(1 - \frac{s}{x}\right)p_n(0) = \frac{s}{x}p.$$

Integrating from 0 to x yields (691). □

Lemma 125. *For $x \le s \le 1$ we have*

$$p_n(s) \ge p + \frac{1-p}{1-x}(s-x). \tag{692}$$

Consequently,

$$\int_x^1 p_n(s)\, ds \ge \frac{(1-x)(p+1)}{2}. \tag{693}$$

Proof. Again by concavity, the graph of p_n lies above the chord joining (x, p) and $(1, 1)$. The equation of this chord is

$$\ell(s) = p + \frac{1-p}{1-x}(s-x), \qquad s \in [x, 1].$$

Therefore $p_n(s) \geq \ell(s)$ on $[x, 1]$. Integrating,

$$\int_x^1 p_n(s)\, ds \geq \int_x^1 \ell(s)\, ds = p(1-x) + \frac{1-p}{1-x} \cdot \frac{(1-x)^2}{2} = \frac{(1-x)(p+1)}{2},$$

which is (693). □

Combining (691) and (693), we obtain

$$M = \int_0^1 p_n(s)\, ds \geq \frac{xp}{2} + \frac{(1-x)(p+1)}{2} = \frac{1+p-x}{2}. \tag{694}$$

Hence

$$x \geq 1 + p - 2M. \tag{695}$$

Since

$$\tau = \frac{1}{M} \int_0^x p_n(s)\, ds, \tag{696}$$

(691) and (695) imply

$$\tau \geq \frac{xp}{2M} \geq \frac{p(1+p-2M)}{2M}. \tag{697}$$

Thus it is enough to prove

$$\frac{p(1+p-2M)}{2M} > \frac{p-M}{2p-1}. \tag{698}$$

After multiplying by the positive quantity $2M(2p-1)$, this is equivalent to

$$G(p, M) = 2p^3 + (1-4M)p^2 - p + 2M^2 > 0. \tag{699}$$

We now check this carefully.

Lemma 126. *If*

$$M \geq \frac{2}{3}, \qquad p \in (M, 1], \tag{700}$$

then

$$G(p, M) = 2p^3 + (1-4M)p^2 - p + 2M^2 > 0. \tag{701}$$

Proof. For fixed M, consider $G(\cdot, M)$ on $[M, 1]$. We have

$$\partial_p G(p, M) = 6p^2 + 2(1-4M)p - 1, \tag{702}$$

$$\partial_{pp} G(p, M) = 12p + 2 - 8M. \tag{703}$$

Since $p \geq M \geq \frac{2}{3}$,

$$\partial_{pp} G(p, M) \geq 12M + 2 - 8M = 4M + 2 > 0. \tag{704}$$

So $G(\cdot, M)$ is strictly convex on $[M, 1]$, hence has a unique minimizer there. We must show that this minimum is positive.

Case 1: $M \geq \frac{7}{8}$**.** Then

$$\partial_p G(1, M) = 7 - 8M \leq 0. \tag{705}$$

Since $\partial_p G(\cdot, M)$ is increasing, it follows that

$$\partial_p G(p, M) \leq 0 \qquad \text{for all } p \in [M, 1]. \tag{706}$$

Hence $G(\cdot, M)$ is decreasing on $[M, 1]$ and its minimum is attained at $p = 1$:

$$G(1, M) = 2(1 - M)^2 > 0. \tag{707}$$

Case 2: $\frac{2}{3} \leq M < \frac{7}{8}$**.** Now

$$\partial_p G(M, M) = -2M^2 + 2M - 1 < 0, \qquad \partial_p G(1, M) = 7 - 8M > 0. \tag{708}$$

Thus the unique minimizer $p_*(M)$ lies in $(M, 1)$ and is characterized by

$$\partial_p G(p_*(M), M) = 0. \tag{709}$$

Solving the quadratic equation gives

$$p_*(M) = \frac{4M - 1 + \sqrt{16M^2 - 8M + 7}}{6}. \tag{710}$$

Define

$$H(M) = G(p_*(M), M). \tag{711}$$

A direct simplification yields

$$H(M) = \frac{-64M^3 + 156M^2 - 48M + 10 - (16M^2 - 8M + 7)^{3/2}}{54}. \tag{712}$$

Differentiating twice one finds

$$H''(M) = -\frac{4}{9\sqrt{16M^2 - 8M + 7}}\left(64M^2 - 32M + 16 + (16M - 13)\sqrt{16M^2 - 8M + 7}\right). \tag{713}$$

For $M \in [\frac{2}{3}, \frac{7}{8}]$ the bracket is strictly positive, hence $H''(M) < 0$. Therefore H is concave on $[\frac{2}{3}, \frac{7}{8}]$, so its minimum on this interval is attained at one of the endpoints.

At $M = \frac{7}{8}$ we have $p_*(\frac{7}{8}) = 1$, hence

$$H\Big(\frac{7}{8}\Big) = G\Big(1, \frac{7}{8}\Big) = \frac{1}{32} > 0. \tag{714}$$

At $M = \frac{2}{3}$ one obtains

$$H\Big(\frac{2}{3}\Big) = \frac{766 - 79\sqrt{79}}{1458} > 0. \tag{715}$$

Thus $H(M) > 0$ on $[\frac{2}{3}, \frac{7}{8}]$.

Combining the two cases proves $G(p, M) > 0$ for all $M \geq \frac{2}{3}$ and $p \in (M, 1]$. □

By (670), the hypotheses of *Lemma 126* are satisfied. Consequently (686) holds whenever $p_n(x) > \frac{1}{2}$.

The branch $p_n(x) < \frac{1}{2}$

We now prove (687). Fix again n and $x \in [0, 1]$, and write

$$p = p_n(x), \qquad M = M_n, \qquad \tau = tau_n(x). \tag{716}$$

Assume

$$p < \frac{1}{2}. \tag{717}$$

We need to show

$$\tau < \frac{M - p}{1 - 2p}. \tag{718}$$

Since A_{n+1} is convex, increasing, and satisfies $A_{n+1}(0) = 0$, $A_{n+1}(1) = 1$, it lies below the diagonal:

$$\tau_n(x) = A_{n+1}(x) \leq x. \tag{719}$$

Also

$$p_n(x) = \eta(A_n^{-1}(x)) \geq \eta(x) = 2x - x^2. \tag{720}$$

Since $\gamma = \eta^{-1}$ is increasing, this implies

$$x \leq \gamma(p) = 1 - \sqrt{1-p}. \tag{721}$$

Hence

$$\tau \leq x \leq \gamma(p). \tag{722}$$

Thus it is enough to prove

$$\gamma(p) < \frac{M-p}{1-2p}. \tag{723}$$

Because $M \geq \frac{2}{3}$, it is enough to prove

$$\gamma(p) < \frac{\frac{2}{3}-p}{1-2p}, \qquad 0 < p < \frac{1}{2}. \tag{724}$$

Set $u = \sqrt{1-p} \in \left(\frac{1}{\sqrt{2}}, 1\right)$. Then $p = 1 - u^2$, and the inequality becomes

$$1 - u < \frac{u^2 - \frac{1}{3}}{2u^2 - 1}. \tag{725}$$

Since $2u^2 - 1 > 0$, this is equivalent to

$$6u^3 - 3u^2 - 3u + 2 > 0. \tag{726}$$

The cubic on the left has derivative

$$18u^2 - 6u - 3, \tag{727}$$

whose unique zero in $\left(\frac{1}{\sqrt{2}}, 1\right)$ is

$$u_* = \frac{1+\sqrt{7}}{6}. \tag{728}$$

A direct evaluation gives

$$6u_*^3 - 3u_*^2 - 3u_* + 2 = \frac{13}{9} - \frac{7\sqrt{7}}{18} > 0. \tag{729}$$

Since the cubic is positive at $u = 1/\sqrt{2}$ and at $u = 1$, it is positive throughout $\left(\frac{1}{\sqrt{2}}, 1\right)$. This proves (687).

Conclusion of the branch analysis and proof of the L_n convergence

We have proved:

Claim 127. *There exists N such that for every $n \geq N$ and every $x \in [0,1]$,*

$$K_n(x) < 1. \tag{730}$$

Equivalently,

$$q_n = \sup_{x\in[0,1]} K_n(x) < 1 \qquad (n \geq N). \tag{731}$$

Proof. The two branch inequalities (686) and (687) show that (684) holds for every x, hence $K_n(x) < 1$. Continuity of K_n on the compact interval $[0,1]$ implies $q_n < 1$. □

We now can formulate the following

Theorem 128. *The sequence $\{L_n\}_{n\geq 0}$ converges uniformly on $[0,1]$.*

Proof. Because $\{L_n\}_{n\geq 0}$ is relatively compact in $C([0,1])$, the transformed sequence $\{R_n\}_{n\geq 0}$ is also relatively compact in $C([0,1])$ by (649). Then, using (663),

$$A_n(x) = R_{n+1}(\eta(x)), \tag{732}$$

so $\{A_n\}_{n\geq 0}$ is relatively compact as well.

Let Ω be the cluster set of $\{A_n\}_{n\geq 0}$ in $C([0,1])$:

$$\Omega = \{A \in C([0,1]) \,:\, \exists n_k \to \infty \text{ with } A_{n_k} \to A \text{ uniformly on } [0,1]\}\,. \tag{733}$$

Then Ω is nonempty and compact.

Step 1: every cluster point is admissible. Since each A_n belongs to $\mathcal{A}$ and the defining properties of $\mathcal{A}$ are preserved under uniform limits, every $A \in \Omega$ belongs to $\mathcal{A}$. In particular, A is continuous, strictly increasing, convex, and satisfies $A(0) = 0$, $A(1) = 1$.

Step 2: continuity of the inverse and of T. Let $A_k \to A$ uniformly in $C([0,1])$, with $A_k, A \in \mathcal{A}$. Since A is strictly increasing and continuous on $[0,1]$, its inverse A^{-1} is continuous. We claim that

$$A_k^{-1} \to A^{-1} \qquad \text{uniformly on } [0,1]. \tag{734}$$

Indeed, fix $\varepsilon > 0$. Since A is strictly increasing and continuous,

$$m_\varepsilon = \min\{|A(u) - A(v)| :\ |u - v| \geq \varepsilon,\ u, v \in [0,1]\} > 0. \tag{735}$$

If $\|A_k - A\|_\infty < m_\varepsilon/2$ and $y \in [0,1]$, let $x = A^{-1}(y)$, $x_k = A_k^{-1}(y)$. If $|x_k - x| \geq \varepsilon$, then by the definition of m_ε,

$$|A(x_k) - A(x)| \geq m_\varepsilon. \tag{736}$$

But

$$|A(x_k) - A(x)| = |A(x_k) - A_k(x_k)| \leq \|A - A_k\|_\infty < \frac{m_\varepsilon}{2}, \tag{737}$$

a contradiction. Hence $|x_k - x| < \varepsilon$ uniformly in y, proving $A_k^{-1} \to A^{-1}$ uniformly.

It follows that the maps

$$A \mapsto q = A^{-1}, \qquad A \mapsto p = \eta \circ q, \qquad A \mapsto M = \int_0^1 p, \qquad A \mapsto \tau = TA \tag{738}$$

are continuous on $\mathcal{A}$ in the sup norm. Consequently, the coefficient

$$K(A, x) = \frac{(1 - \tau(x))p(x) + \tau(x)(1 - p(x))}{M} \tag{739}$$

is jointly continuous in $(A, x) \in \mathcal{A} \times [0,1]$.

Step 3: index-free form of the branch analysis. Let $A \in \Omega$. Define

$$q = A^{-1}, \qquad p = \eta(q), \qquad M = \int_0^1 p, \qquad \tau = TA. \tag{740}$$

Exactly the same arguments as in the indexed proof apply: p is increasing and concave, $M \geq 2/3$, and the two branch inequalities proved above show that

$$K(A, x) < 1 \qquad \text{for every } x \in [0,1]. \tag{741}$$

Thus

$$K(A, x) < 1 \qquad \forall (A, x) \in \Omega \times [0,1]. \tag{742}$$

Step 4: compactness yields a uniform gap. Since $\Omega \times [0,1]$ is compact and K is continuous, the maximum

$$q_* = \max_{A\in\Omega} \max_{x\in[0,1]} K(A, x) \tag{743}$$

is attained. Because $K(A, x) < 1$ everywhere on $\Omega \times [0, 1]$, we have

$$q_* < 1. \tag{744}$$

By continuity of K, there exists an open neighborhood $\mathcal{U}$ of Ω in $C([0, 1])$ such that

$$\sup_{A \in \mathcal{U}} \sup_{x \in [0,1]} K(A, x) \le q \tag{745}$$

for some $q \in (q_*, 1)$.

Step 5: the orbit eventually enters the contraction neighborhood. We claim that there exists N_0 such that

$$A_n \in \mathcal{U} \qquad \text{for all } n \ge N_0. \tag{746}$$

If not, there would exist a subsequence $A_{n_k} \notin \mathcal{U}$. By relative compactness, this subsequence has a uniformly convergent subsubsequence with limit in Ω, contradicting the fact that $\mathcal{U}$ is a neighborhood of Ω.

Step 6: tail contraction and convergence. For $n \ge N_0$, the segment joining A_n and A_{n+1} lies in a convex neighborhood on which the derivative bound is at most $q < 1$. By the *Banach-space mean-value theorem*,

$$\|A_{n+2} - A_{n+1}\|_\infty = \|T(A_{n+1}) - T(A_n)\|_\infty \le q\|A_{n+1} - A_n\|_\infty. \tag{747}$$

By iteration,

$$\|A_{n+k+1} - A_{n+k}\|_\infty \le q^k \|A_{n+1} - A_n\|_\infty, \qquad k \ge 0. \tag{748}$$

Summing the geometric series shows that $\{A_n\}_{n \ge 0}$ is *Cauchy* in $C([0, 1])$, hence converges uniformly to some $A_\infty \in C([0, 1])$.

Now (663) implies

$$R_{n+1}(x) = A_n(\gamma(x)) \longrightarrow A_\infty(\gamma(x)) = R_\infty(x) \tag{749}$$

uniformly on $[0, 1]$. Finally, *Lemma 121* and the symmetry-gap convergence (646) imply that

$$L_n \longrightarrow L_\infty \tag{750}$$

uniformly on $[0, 1]$.

This completes the proof. □

Remark 129. Alternatively, if we take a closer look at the preceeding proofs, we can reason in the following way.

By Claim 127, for every $n \ge N$,

$$q_n = \sup_{x \in [0,1]} K_n(x) < 1. \tag{751}$$

The preceding algebra shows that equality in the two branch inequalities is excluded throughout the tail. In particular, the tail coefficients cannot accumulate at 1. Hence there exists $q \in (0, 1)$ and an index, still denoted by N, such that

$$q_n \le q \qquad \text{for all } n \ge N. \tag{752}$$

Now the Banach-space mean-value theorem and the contraction principle imply that the tail map is uniformly contractive in the sup norm. Therefore $\{A_n\}_{n \ge 0}$ is a Cauchy sequence in $C([0, 1])$ and converges uniformly to a limit A_∞. By (663),

$$R_{n+1}(x) = A_n(\gamma(x)) \longrightarrow A_\infty(\gamma(x)) = R_\infty(x) \tag{753}$$

uniformly on $[0, 1]$. Finally, Lemma 121 and the symmetry-gap convergence (646) yield

$$L_n \longrightarrow L_\infty$$

uniformly on $[0, 1]$, where L_∞ is the symmetric function reconstructed from R_∞ by (652).

Final conclusion for the *Fréchet-Hoeffding lower-bound* iteration

We now turn to an advisable but yet not crucial for our analysis final objective: identifying possible closed-form expressions for the limit of $\{L_n\}_{n\geq 0}$, a task clearly achieved for the *upper-bound* case in the forthcoming *Appendix C.2.*

We have the following

Lemma 130. *The iterative equation (222) simplifies to a form of the iterative equation (45) from [31].*

Proof. Before going to the proof of the proposition, we start with some simplifications. Substitute a new function $H^n(x) : [-\frac{1}{2}, \frac{1}{2}] \to [0,1]$ such that $H^{n,-1}(x) = \frac{1}{2} - L^{n,-1}(x)$. Since a direct differentiation in (222) gives that $L^n(x)$ is monotonous, and thus also $H^n(x)$ and $H^{n,-1}(x)$, inverting the latter, gives $L^n(x) = H^n(\frac{1}{2} - x)$. Now we can get for (222) in terms of H^n

$$H^{n+1}(\frac{1}{2} - x) = \frac{\int_0^x \left[\frac{1}{4} - \left(H^{n,-1}(u)\right)^2\right] du}{\int_0^1 \left[\frac{1}{4} - \left(H^{n,-1}(u)\right)^2\right] du}, \text{ with } H^0(\frac{1}{2} - x) = \frac{\int_0^x F^{-1}(u)\left(1 - F^{-1}(u)\right) du}{\int_0^1 F^{-1}(u)\left(1 - F^{-1}(u)\right) du}. \tag{754}$$

Next, to simplify further, we would like to substitute a new function $K^{n,-1}(x)$ for the integrand $\frac{1}{4} - \left(H^{n,-1}(x)\right)^2$. We consider two cases based on the monotonicity of the function $\frac{1}{4} - \left(H^{n,-1}(x)\right)^2$, so that we can take its proper inverse and thus find $K^n(x)$. That would allow to effectively see what new equation (754) gets transformed to. A right control of the domain and range of $K^{n,-1}(x)$ is needed.

In the first case, substitute initially a new function $K_1^{n,-1}(x)$ such that $K_1^{n,-1}(x) = \frac{1}{4} - \left(H^{n,-1}(x)\right)^2$ and require $K_1^{n,-1}(x)$ be decreasing. This monotonicity is valid only for $H^{n,-1}(x) \in [0, \frac{1}{2}]$. For the latter interval, we get that it holds $K_1^{n,-1}(x) \in [0, \frac{1}{4}]$. Since $H^{n,-1}(x)$ is decreasing ($H^n(x) = L^n(\frac{1}{2} - x)$) and $H^{n,-1}(x) = \frac{1}{2}$ for $x = 0$, for the range $[0, \frac{1}{2}]$ of $H^{n,-1}(x)$ we get the domain $x \in [0, H^n(0)]$. Note that since we do not know for which x it holds $H^{n,-1}(x) = 0$, we have put in the domain the point $H^n(0)$[1]. In terms of $K_1^{n,-1}(x)$, $H^{n,-1}(x) = 0$ implies $K_1^{n,-1}(x) = \frac{1}{4}$ and so for the domain of $K_1^{n,-1}(x)$ we get $x \in [0, K_1^n(\frac{1}{4})]$. Again, since we do not know for which x it holds $K_1^{n,-1}(x) = \frac{1}{4}$, we have put in the domain $K_1^n(\frac{1}{4})$. The derivations imply that effectively we can define $K_1^{n,-1}(x) : [0, K_1^n(\frac{1}{4})] \to [0, \frac{1}{4}]$,or equivalently $K_1^n(x) : [0, \frac{1}{4}] \to [0, K_1^n(\frac{1}{4})]$, with $K_1^{n,-1}(x) = \frac{1}{4} - \left(H^{n,-1}(x)\right)^2$. Additionally, for $x \in [0, K_1^n(\frac{1}{4})]$ we have $H^{n,-1}(x) = \sqrt{\frac{1}{4} - K_1^{n,-1}(x)}$. If we take inverses in the latter and consider the range of $\sqrt{\frac{1}{4} - K_1^{n,-1}(x)}$, for $x \in [0, \frac{1}{2}]$ we have $H^n(x) = K_1^n(\frac{1}{4} - x^2)$ as well as for $x \in [0, \frac{1}{4}]$ we have $K_1^n(x) = H^n\left(\sqrt{\frac{1}{4} - x}\right)$.

In the second case, substitute initially a new function $K_2^{n,-1}(x)$ such that $K_2^{n,-1}(x) = \frac{1}{4} - \left(H^{n,-1}(x)\right)^2$ and require $K_2^{n,-1}(x)$ be increasing. This monotonicity is valid only for $H^{n,-1}(x) \in [-\frac{1}{2}, 0]$. For the latter interval we get that again it holds $K_2^{n,-1}(x) \in [0, \frac{1}{4}]$. Since $H^{n,-1}(x)$ is decreasing ($H^n(x) = L^n(\frac{1}{2} - x)$) and $H^{n,-1}(x) = -\frac{1}{2}$ for $x = 1$, for the range $[-\frac{1}{2}, 0]$ of $H^{n,-1}(x)$ we get the domain $x \in [H^n(0), 1]$. Since we do not know for which x it holds $H^{n,-1}(x) = 0$, we have put in the domain the point $H^n(0)$. In terms of $K_2^{n,-1}(x)$, $H^{n,-1}(x) = 0$ again implies $K_2^{n,-1}(x) = \frac{1}{4}$ and so for the domain of $K_2^{n,-1}(x)$ we get $x \in [K_2^n(\frac{1}{4}), 1]$. Again, since we do not know for which x it holds $K_2^{n,-1}(x) = \frac{1}{4}$, we have put in the domain $K_2^n(\frac{1}{4})$. The derivations imply that effectively we can define $K_2^{n,-1}(x) : [K_2^n(\frac{1}{4}), 1] \to [0, \frac{1}{4}]$,or equivalently $K_2^n(x) : [0, \frac{1}{4}] \to [K_2^n(\frac{1}{4}), 1]$, with $K_2^{n,-1}(x) = \frac{1}{4} - \left(H^{n,-1}(x)\right)^2$. Additionally, for $x \in [K_2^n(\frac{1}{4}), 1]$ we have $H^{n,-1}(x) = -\sqrt{\frac{1}{4} - K_2^{n,-1}(x)}$. If we take inverses in the latter and consider the range of $\sqrt{\frac{1}{4} - K_2^{n,-1}(x)}$, for $x \in [-\frac{1}{2}, 0]$ we have $H^n(x) = K_2^n(\frac{1}{4} - x^2)$ as well as for $x \in [0, \frac{1}{4}]$ we have $K_2^n(x) = H^n\left(-\sqrt{\frac{1}{4} - x}\right)$.

Now, we can rewrite (754) in terms of $K^n(.)$. Before doing this, we should note that effectively we have distinguished between the two cases above based on the integrand in (754). Within the process, we specially considered the range and domain of $H^n(.)$ and $K^n(.)$ and of their inverses. All that was done up to iteration

1 For $H^n(.)$, respectively $L^n(.)$, the only known points, independent from the starting distribution $F(.)$, are derived by the defining equations (222) and (754) and they are 0 and 1 for $L^n(.)$ and $\pm\frac{1}{2}$ for $H^n(.)$. More precisely, we have $L^n(0) = 0$, $L^n(1) = 1$ and $H^n\left(-\frac{1}{2}\right) = 1$, $H^n\left(\frac{1}{2}\right) = 0$.

n. Moving to iteration $n+1$, however, we have to be careful. The range and domain of $H^{n+1}(.)$ and $K^{n+1}(.)$ on the left hand side of (754) have to be consistent with their counterparts on the right hand side of the equation coming from the integrand in which inverses of $H^n(.)$ and $K^n(.)$ participate.

We consider two new cases based on the order of $\frac{1}{2}$ and $H^n(0)$: 1) $\frac{1}{2} \leq H^n(0)$ (with $H^n(0) = K_1^n(\frac{1}{4}) = K_2^n(\frac{1}{4}) = L^n(\frac{1}{2})$ clearly being valid as well). Three sub-cases occur based on the domain: a) $x \in [0, \frac{1}{2}]$, b) $x \in [\frac{1}{2}, H^n(0)]$, and c) $x \in [H^n(0), 1]$; and 2) $H^n(0) \leq \frac{1}{2}$ (with $H^n(0) = K_1^n(\frac{1}{4}) = K_2^n(\frac{1}{4}) = L^n(\frac{1}{2})$ clearly being valid as well). Three sub-cases occur again: a) $x \in [0, H^n(0)]$, b) $x \in [H^n(0), \frac{1}{2}]$, and c) $x \in [\frac{1}{2}, 1]$. Within each of them, we shall consider the range consistency. For space consideration, we won't show these details and we will go directly to the next step.

In that skipped analysis, it turns out that we are allowed not to differentiate explicitly between $K_1(.)$ and $K_2(.)$ in each of the two cases above across their sub-cases. This is due to the fact that we can take the union of the the two cases and their sub-cases in terms of the domain both of $K(.)$ (iteration $n+1$) and $K^{-1}(.)$ (iteration n). Since the latter was the distinguishing feature for the sub-cases, i.e. the subscripts usage, writing the equations in simpler uniform way is possible

$$K^{n+1}\left(x(1-x)\right) = \left\{ \frac{\int_0^x K^{n,-1}(u)du}{\int_0^1 K^{n,-1}(u)du}, K^0(x(1-x)) = \frac{\int_0^x F^{-1}(u)\left(1-F^{-1}(u)\right)du}{\int_0^1 F^{-1}(u)\left(1-F^{-1}(u)\right)du}, x \in [0,1]. \right. \tag{755}$$

Effectively we had the substitutions

$$\begin{aligned} L^n(\frac{1}{2}-x) &= H^n(x) = K^n(\frac{1}{4}-x^2), x \in [-\frac{1}{2}, \frac{1}{2}] && (756) \\ L^n(x) &= K^n(x(1-x)), x \in [0,1]. && (757) \end{aligned}$$

We can further note that there seems to be a shortcut approach for going directly from (754) to (755) by just posing initially the (unindexed) $K^{n,-1}(x) : [0,1] \to [0, \frac{1}{4}]$ by $K^{n,-1}(x) = \frac{1}{4} - \left(H^{n,-1}(x)\right)^2$. Then we invert two times[2] and get $H^n(x) = K^n(\frac{1}{4} - x^2)$ and from the latter also $L^n(x) = H^n(\frac{1}{2} - x) = K^n(\frac{1}{4} - (\frac{1}{2} - x)^2) = K^n(x(1-x))$. Yet, this approach is only heuristic, since it does not consider well the domain and range effects we discussed. Ignoring them is inappropriate since there could appear impossible cases and the union argument above not to work as fluently as it did.

We proceed now with (755). Based on the monotonicity of the function $x \longmapsto x(1-x)$ on $x \in [0,1]$, we have two cases

$$\begin{aligned} L^n(x) &= K_*^n(x(1-x)), x \in [0, \frac{1}{2}] && (758) \\ L^n(x) &= K_{**}^n(x(1-x)), x \in [\frac{1}{2}, 1], && (759) \end{aligned}$$

where on $x \in [0, \frac{1}{4}]$ holds

$$K_*^{n+1}(x) = \frac{\int_0^{\varphi_*(x)} K_*^{n,-1}(u)du}{\underbrace{\int_0^{K_*^n(\frac{1}{4})} K_*^{n,-1}(u)du + \int_{K_*^n(\frac{1}{4})}^1 K_{**}^{n,-1}(u)du}_{\mu^*}}, K_*^0(x) = \frac{\int_0^{\varphi_*(x)} F^{-1}(u)\left(1-F^{-1}(u)\right)du}{\int_0^1 F^{-1}(u)\left(1-F^{-1}(u)\right)du} \tag{760}$$

$$K_{**}^{n+1}(x) = \frac{\int_0^{\varphi_{**}(x)} K_{**}^{n,-1}(u)du}{\underbrace{\int_0^{K_{**}^n(\frac{1}{4})} K_*^{n,-1}(u)du + \int_{K_{**}^n(\frac{1}{4})}^1 K_{**}^{n,-1}(u)du}_{\mu^{**}}}, K_{**}^0(x) = \frac{\int_0^{\varphi_{**}(x)} F^{-1}(u)\left(1-F^{-1}(u)\right)du}{\int_0^1 F^{-1}(u)\left(1-F^{-1}(u)\right)du} \tag{761}$$

2 First the compound function $\left(H^{n,-1}(x)\right)^2$ and second $H^{n,-1}(x)$ intself. In this way, we do not need to specify whether we have $H^{n,-1}(x) = -\sqrt{\frac{1}{4} - K^{n,-1}(x)}$ or $H^{n,-1}(x) = \sqrt{\frac{1}{4} - K^{n,-1}(x)}$ at the first step, since somehow the effect is cancelled out in the second one.

We used that for $x \in [0, \frac{1}{2}]$, the function $x(1-x)$ is monotonically increasing, and for $x \in [\frac{1}{2}, 1]$ it is monotonically decreasing, which allows us to invert it. This gives rise to the functions: $\varphi_*(x) : [0, \frac{1}{4}] \to [0, \frac{1}{2}]$ satisfying $\varphi_*(x) = \frac{1-\sqrt{1-4x}}{2}$ and $\varphi_{**}(x) : [0, \frac{1}{4}] \to [\frac{1}{2}, 1]$ satisfying $\varphi_{**}(x) = \frac{1+\sqrt{1-4x}}{2}$. Effectively, we had the substitutions

$$K_*^n(x) = L^n\left(\frac{1-\sqrt{1-4x}}{2}\right), x \in [0, \frac{1}{4}] \tag{762}$$

$$K_{**}^n(x) = L^n\left(\frac{1+\sqrt{1-4x}}{2}\right), x \in [0, \frac{1}{4}]. \tag{763}$$

Additionally, we have the special points: $K_*^n(0) = 0, K_{**}^n(0) = 1$, and $K_*^n(\frac{1}{4}) = K_{**}^n(\frac{1}{4}) = L^n(\frac{1}{2})$.

Equations (760) and (761) allow to find $L^n(x)$ for $x \in [0, 1]$ by uniting the two cases. We have to solve each of the equations (760) and (761) in a stand-alone way at its domain and find the $K_*^n(x)$ and $K_{**}^n(x)$. Then for $x \in [0, \frac{1}{2}]$, we plug $x(1-x)$ into $K_*^n(x)$ to find $L^n(x) = K_*^n(x(1-x))$. For $x \in [\frac{1}{2}, 1]$, we plug $x(1-x)$ into $K_{**}^n(x)$ to find $L^n(x) = K_{**}^n(x(1-x))$.

We can observe that (760) and (761) share a similar functional form with the only difference coming from the respective domains and the upper integral limit functions. Additionally, we can notice the similarity of the two equations to equation (45), i.e., the main equation handled in [31]. Here the upper integral limits are the functions $\varphi_*(x)$ and $\varphi_{**}(x)$, while in the aforementioned cases is just x. ☐

Remark 131. Even in the simplified form provided by *Lemma 130*, obtaining a closed-form expression for either the limit of *(222)* or the solution to the functional equation *(267)* remains a non-trivial task.

Remark 132. Differentiating (760) and (761) and then substituting $h^*(x) : [0, \frac{1}{4}] \longrightarrow [0, \frac{1}{4}]$ such that $h^*(x) = K_*^{n,-1}(\varphi_*(x))$, leads to the following functional equation

$$(h^*)'(x) = \frac{\mu^*}{h^*(h^*(x))} \frac{\varphi_*'(x)}{\varphi_*'(h^*(x))}, \tag{764}$$

with $h^*(0) = 0$, $h^*(\frac{1}{4}) = \frac{1}{4}$, and $(h^*)'(\frac{1}{4}) = 4\mu^*$.

Analogously by substituting $h^{**}(x) : [0, \frac{1}{4}] \longrightarrow [\frac{1}{4}, 1]$ such that $h^{**}(x) = K_{**}^{n,-1}(\varphi_*(x))$ we get

$$(h^{**})'(x) = \frac{\mu^{**}}{h^{**}(h^{**}(x))} \frac{\varphi_{**}'(x)}{\varphi_{**}'(h^{**}(x))}, \tag{765}$$

with $h^{**}(0) = 1$, $h^{**}(\frac{1}{4}) = \frac{1}{4}$, and $(h^{**})'(\frac{1}{4}) = 4\mu^{**}$.

Under further regularity/dynamical conditions (e.g., h^* and h^{**} analytic on $(0, \frac{1}{4}]$ (see [8]), increasing on $(0, \frac{1}{4}]$, or $\frac{1}{4}$ being attracting, the two functional–differential equations typically become well-posed existence/uniqueness problems. Yet, even the methods in [8] do not provide straightforward path to find a closed form solution.

Example 133. The ansatz function $G_-(x)$, defined by

$$G_-(x) = \begin{cases} \frac{1-\sqrt{1-[4x(1-x)]^{\frac{\sqrt{5}+1}{2}}}}{2}, 0 \le x \le \frac{1}{2} \\ \frac{1+\sqrt{1-[4x(1-x)]^{\frac{\sqrt{5}+1}{2}}}}{2}, \frac{1}{2} < x \le 1 \\ 0, x < 0 \\ 1, x > 1 \end{cases} \tag{766}$$

provides a strong approximation for the limiting analytic function. This conclusion is based on heuristic reasoning, supported by empirical experiments, that involves approximating the terms $\varphi_*'(x)$, $\varphi_{**}'(x)$, $\varphi_*'(h^*(x))$, and $\varphi_{**}'(h^{**}(x))$ and the ratios participating in *(764)* and *(765)*.

We conclude the appendix with several plots illustrating the dynamics and providing further visual intuition. We take as in *Section 4.3* $F_1 \sim$ *Lognormal*$(0.2, 0.5)$. *Figures C1.1-2* plot the evolution of the d.f.s of the marginals and that of the compounded inverses. They are counterparts to the plots of Section 4.3.

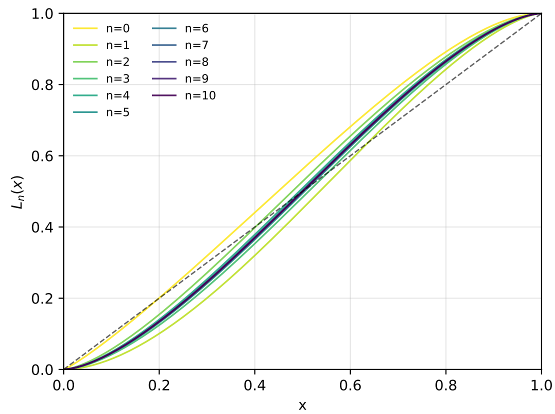

Figure C1.1:
$L^n(x)$ evolution

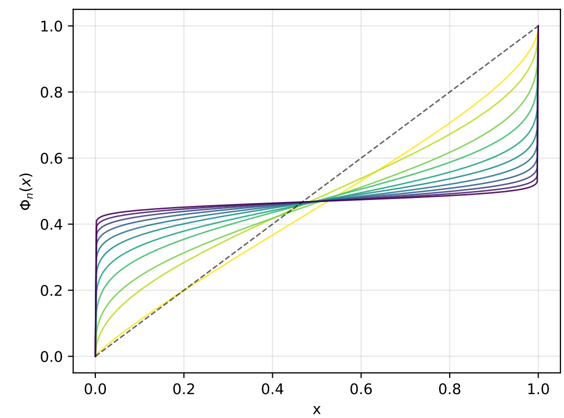

Figure C1.2:
Compounds of the inverse marginal d.f.s - $\Phi_n(x)$

We can clearly see the difference. The crossings do not tend monotonically to the right as the case of RR_2 density was, they rather oscillate towards 0.5. Exactly it is expected to be the limit of $\Phi_n(x)$ as we discussed.

Appendix C.2

Hereinafter, our focus is on the iterative equation (62). For convenience we denote below any of the marginal distributions $L^n_{1+}(x)$ or $L^n_{2+}(x)$ by $L_n(x)$ since the properties proved hold for both of them. Thus, our setting becomes:

Claim 134. *Keeping to the posed assumptions in the main text, for the iterative map*

$$L^{n+1}(x) = \frac{\int_0^x \left(L^{n,-1}(u)\right)^2 du}{\int_0^1 \left(L^{n,-1}(u)\right)^2 du}, \quad \textit{with } L^0(x) = \frac{\int_0^x \left(F^{-1}(u)\right)^2 du}{\int_0^1 \left(F^{-1}(u)\right)^2 du} \tag{767}$$

a uniform convergence of $L^n(x)$ holds towards the distribution function $G_+(x)$

$$G_+(x) = \begin{cases} x^2, 0 \le x \le 1 \\ 0, x < 0 \\ 1, x > 1. \end{cases} . \tag{768}$$

Proof. We can do the proof by induction verbally following the logic of [31]. For space considerations it won't be presented. We will just point out that again we can find majorizing polynomials such for $0 \le x \le 1$ holds

$$\begin{aligned} 0 &\le F(x) \le 1 \\ 0 &\le L_0(x) \le x \\ x^3 &\le L_1(x) \le x \\ x^3 &\le L_2(x) \le x^{\frac{5}{3}} \\ &\dots \\ x^{\alpha_{n+1}} &\le L_n(x) \le x^{\alpha_n}, n \text{ - } odd \\ x^{\alpha_n} &\le L_n(x) \le x^{\alpha_{n+1}}, n \text{ - } even, \end{aligned} \tag{769}$$

where the sequence $\alpha_1 = 1$, $\alpha_2 = 3$, $\alpha_3 = \frac{5}{3}$,... is given recursively by:

$$\alpha_{n+1} = 1 + \frac{2}{\alpha_n}. \tag{770}$$

The sequence (770) is convergent (bounded and monotonous in even and odd members, sharing a joint candidate limit). Solving the equation $\alpha = 1 + \frac{2}{\alpha}$ gives us:

$$\lim_{n\to+\infty} L_n(x) = x^2 \text{ for } 0 \le x \le 1. \tag{771}$$

The point-wise convergence of the sequence of functions $L_n(x)$ in the unit interval (compact set), together with their continuity and monotonicity in x, makes the continuity of the limiting function a necessary and sufficient condition for uniform convergence in the unit interval. Thus the sequence of functions $L_n(x)$ are not only point-wise convergent but also uniformly convergent in $[0, 1]$. □

Remark 135. For the *Fréchet-Hoeffding* upper-bound, we find that $L_n(x)$ has no crossing points in (0,1). This follows from the subdiagonal pattern established by the polynomial majorization from *Claim* 134 and is in contrast to the behavior observed for the lower-bound case in *Appendix C.1.*

Lemma 136. *Under the Fréchet-Hoeffding upper-bound iteration (768), the compound maps*

$$\Phi_n = T_0 \circ T_1 \circ \cdots \circ T_n, \tag{772}$$

where again T_n denotes the inverse of $L_n(x)$, converge pointwise and locally uniformly on $(0, 1)$ to the constant map

$$\Phi_\infty(x) = 1 \qquad x \in (0, 1). \tag{773}$$

Proof. From (769) follows

$$\begin{aligned} x^{\frac{1}{\alpha_n}} &\le T_n(x) \le x^{\frac{1}{\alpha_{n+1}}}, n \text{ - } odd \\ x^{\frac{1}{\alpha_{n+1}}} &\le T_n(x) \le x^{\frac{1}{\alpha_n}}, n \text{ - } even. \end{aligned} \tag{774}$$

We may note that all powers of x are less than 1. Therefore

$$\Phi_n = T_0 \circ T_1 \circ \cdots \circ T_n \ge \prod_{j=0}^{n} x^{\frac{1}{\gamma_j}} = x^{\varepsilon^n} \to 1, \tag{775}$$

where $\exists\ \varepsilon$ and γ_j such that for $\forall\ j$ holds $\alpha_j < \gamma_j < \varepsilon \in (0, 1)$. Additionally, $T^n(x) \le 1$ for all n and thus $\Phi_n(x) \le 1$ as well. The squeeze gives the result of the theorem. □

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